Enhancing reliability in classification problems using traditional machine learning and meta-learning models

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Abstract

This study aims to address critical gaps in machine learning (ML) reliability and transparency by developing and rigorously evaluating a novel, modality-sensitive statistical classification framework for reliability enhancement. Adopting a dual data-centric and model-centric approach, the research will systematically benchmark the influence of various data partitioning strategies on model generalisation and statistical stability across structured, text, and image datasets.

Following this, the framework will implement and compare the performance of both traditional ML and meta-learning models to determine their efficacy and adaptability in different data contexts. A key component of this research is the integration of a unified reliability pipeline, which conducts a multi-faceted evaluation encompassing statistical drift detection, bias and fairness auditing, and model explainability. The expected outcome is a statistically sound, reproducible, and ethically aware framework that provides a robust pathway for building trustworthy ML systems, offering significant implications for practice and real-world decision-making.

Keywords: Optimal Data Splitting; Classification ML; Meta-Learning Strategies; Predictive Reliability Evaluation; Model Explainability; Beyond Accuracy-Metrics.

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Table of Abbreviations

Table 1: A table containing a list of abbreviations that will be used throughout the text.

Accuracy	Classification Performance Metric
ANOVA	Analysis of Variance for Statistical Model Comparison
ATTP	Adaptive Train-Test Partitioning
AUC	Area Under the Curve
Bellman Equation	Q-Value Update Formula
Chi-Square Test	Categorical Feature Comparison
CNN	Convolutional Neural Network
Concept Drift	Feature-Target Relationship Shift Over Time
Confusion Matrix	Tabular Representation of Classification Outcomes
Covariate Shift	Distributional Shift in Input Variables
CV	Cross-Validation
Data Drift	Distributional Change in Training vs. Test Sets
Drift Score Index (DSI)	Quantification of Statistical Drift
Equalized Odds	Metric for Evaluating Predictive Fairness
F1-score	Harmonic Mean of Precision and Recall
Feature Drift	Variation in Feature Importance
Group K-Fold CV	Group K-Fold Cross-Validation
Group ShuffleSplit	Group Shuffle-Split Partitioning
i.i.d.	Independently and Identically Distributed
Iterative Optimization	Dynamic Train-Test Refinement Process
Jensen-Shannon (JS) Divergence	Smoothed KL Divergence Variation
K-Fold CV	K-Fold Cross-Validation
k-Means	k-Means Clustering Algorithm
Kolmogorov-Smirnov (K-S) Test	Statistical Test for Distribution Differences
Kullback-Leibler (KL) Divergence	Measure of Distribution Differences
LDA	Linear Discriminant Analysis
Leave-P-Out CV	Leave-P-Out Cross-Validation
LIME	Local Interpretable Model-Agnostic Explanations
Logistic Regression (LR)	Statistical Model for Binary Classification
LSTM	Long Short-Term Memory Model
MAML	Model-Agnostic Meta-Learning
MANOVA	Multivariate Analysis of Variance
Matthews Correlation (MCC)	Classification Performance Metric
Naïve Bayes (NB)	Probabilistic Classifier for Text Classification
Non-Parametric Bootstrap	Model Stability Estimation via Resampling
PCA	Principal Component Analysis
Predictive Parity	Measures Consistent Predictive Accuracy Across Groups
Precision	Classification Metric for Positive Predictions
ProtoNet	Prototype-Based Few-Shot Learning Model
Q-Learning	Reinforcement Learning Algorithm for Partitioning
Random Forest (RF)	Ensemble Learning Algorithm Using Decision Trees
Recall	Classification Metric for True Positive Recovery

Repeated K-Fold CV	Repeated K-Fold Cross-Validation
Repeated Stratified K-Fold CV	Repeated Stratified K-Fold Cross-Validation
ResNet	Residual Neural Network Architecture
Reptile	Gradient-Based Meta-Learning Algorithm
RNN	Recurrent Neural Network
RLAPT-Q Learning	Reinforcement Learning-Based Partitioning - Q Learning
SHAP	SHapley Additive exPlanations for Feature Importance
SVM	Support Vector Machine
Statistical Parity	Fairness Measure for Equal Selection Probability
Stratified Group K-Fold CV	Stratified Group K-Fold Cross-Validation
Stratified K-Fold CV	Stratified K-Fold Cross-Validation
Stratified Sampling	Partitioning Method with Class Proportion Preservation
Stratified ShuffleSplit	Stratified Shuffle-Split Partitioning
Wasserstein Distance (WD)	Metric for Optimal Transport Between Distributions

0.1 Introduction

Machine learning (ML) models are now widely used in many fields, significantly changing how we make decisions (Sharma et al., 2020; Li et al., 2021). In **classification problems**, which involve predicting categories (like diagnosing diseases, detecting fraud, or recognising images), ML algorithms are particularly powerful at finding patterns and making predictions. Because these models are so widely applied, it's increasingly important that they work consistently and reliably in real-world situations.

The long-term success of ML models isn't just about their initial performance; it also depends heavily on their sustained reliability. To clarify this, it's essential to distinguish **reliability** from other common performance metrics like **accuracy**. In machine learning, accuracy is a primary and often reported metric that quantifies how well a model performs at a single point in time. It's typically defined as the proportion of correctly predicted instances out of the total predictions made (e.g., Tharwat, 2018):

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where TP (True Positives), TN (True Negatives), FP (False Positives), and FN (False Negatives) are standard classification outcomes.

While high accuracy is undoubtedly desirable for initial validation, reliability represents a more profound and dynamic aspect of a model's trustworthiness. Generally, reliability refers to the ability of a system or component to perform its required functions under stated conditions for a specified period of time. In the context of computational and machine learning systems, this is extended to include the consistent performance and maintenance of predicted results over time and across varying, yet statistically similar, conditions or data shifts (Verma et al., 2022). In the context of this study, reliability specifically means the model's capacity to maintain stable and consistent performance when faced with new, unseen data or changing operational conditions after it has been trained and put into use. This goes beyond a single accuracy score; it can be conceptually represented as ensuring a low variance in performance metrics (including, but not limited to, accuracy) over time and across diverse, representative unseen data subsets, formally aiming for:

$$\text{Reliability}(M) \propto \frac{1}{\text{Var}(\text{Performance}(M, \mathcal{D}_{\text{unseen}}))} \tag{2}$$

where M is the model and $\mathcal{D}_{\text{unseen}}$ represents new, unseen data. Therefore, a reliable classification model ensures not only consistently accurate predictions but also exhibits

robustness against errors, fairness across different demographic groups, and interpretable decisions that remain understandable over time. This sustained, predictable behavior, distinct from mere snapshot accuracy, is crucial for important, high-stakes applications like those in healthcare or finance (Gama et al., 2014).

0.1.1 Key Determinants of ML Model Reliability

Ensuring the long-term reliability of an ML classification model in real-world applications is a complex endeavor that depends on a synergistic combination of factors. These critical determinants span from the foundational handling of data to the sophistication of model construction and, finally, to the continuous, multi-dimensional evaluation of its performance. Specifically, achieving robust reliability necessitates careful consideration of: data preparation and partitioning, robust model building (including advanced learning paradigms), and a comprehensive reliability analysis and evaluation framework.

Several critical factors can significantly affect how reliable an ML classification model remains once it's in use, necessitating a multi-faceted assessment approach. One major factor is the data partitioning method used. This involves how the data is split into training and testing sets, a fundamental step for evaluating a model's ability to generalize to new data. For instance, a Random Split partitions the full dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$ into training and testing subsets based on a fraction θ:

$$\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}} \sim \text{RandomSplit}(\mathcal{D}, \theta) \tag{3}$$

while a Stratified Split ensures that class proportions are preserved across the subsets:

$$\frac{|\mathcal{D}^c_{\text{train}}|}{|\mathcal{D}_{\text{train}}|} \approx \frac{|\mathcal{D}^c|}{|\mathcal{D}|} \quad \text{and} \quad \frac{|\mathcal{D}^c_{\text{test}}|}{|\mathcal{D}_{\text{test}}|} \approx \frac{|\mathcal{D}^c|}{|\mathcal{D}|} \tag{4}$$

where $\mathcal{D}^c$ denotes instances of class c (Yao et al., 2021). More advanced statistical techniques are often needed to get unbiased performance estimates and prevent issues like **data leakage** (where information from the test set accidentally contaminates the training set) or overly optimistic evaluations (Tan et al., 2021; Gonzalez et al., 2022). These problems can severely undermine a model's trustworthiness, especially in sensitive areas like fairness-critical or high-stakes domains. Furthermore, the complexity of different data modalities—such as structured tabular data (demanding careful scaling and encoding) (Yoo et al., 2022), unstructured text (needing advanced text representations or embeddings) (Wang et al., 2020), and high-resolution image data (using complex convolutional networks) (Khan et al., 2021)—highlights the need for specific approaches tailored to each data type for the prediction of unseen cases, contributing significantly to the "data preparation and cleaning" aspect of reliability.

Beyond how data is handled, optimal model building, encompassing both Traditional ML and Meta-Learning paradigms, plays a pivotal role in achieving reliability. Traditional ML models broadly include statistical, probabilistic, and neural network approaches (such as Logistic Regression, Naive Bayes, Convolutional Neural Networks, Long Short-Term

Memory with Classification-Head networks), along with various ensemble techniques (like Random Forests, Gradient Boosting Trees, Support Vector Machines) that combine multiple learners to improve predictive power (Friedman, 2001). For instance, an ensemble model's final prediction $\hat{y}$ can be an aggregation of individual base model predictions $T_i(X)$:

$$\hat{y} = \frac{1}{M} \sum_{i=1}^{M} T_i(X) \tag{5}$$

where M is the number of base models and X is the input (Breiman, 2001). Alongside these, meta-learning, often described as "learning to learn," has emerged as a powerful paradigm (Hospedales et al., 2020; Wang et al., 2021). This methodology encompasses approaches that enable models to quickly adapt to new tasks or evolving data distributions with minimal new data, thereby promising a new level of adaptability and generalisation crucial for dynamic real-world scenarios and efficient predictive estimates. A core meta-learning objective is to learn initial parameters θ that can quickly adapt to new tasks $\mathcal{T}$, often by optimising for rapid fine-tuning:

$$\min_{\theta} \mathbb{E}_{\mathcal{T} \sim p(\mathcal{T})} [\mathcal{L}_{\mathcal{T}}(f_{\theta'}(\cdot))] \quad \text{where } \theta' = \text{Adapt}(\theta, \mathcal{D}^{\mathcal{T}}_{\text{train}}) \tag{6}$$

This represents learning an adaptation process rather than just a fixed model (Finn et al., 2017). Specific meta-learning techniques include Reptile, Stacking Meta-Learner, and Performance-Aware Meta-Selector.

Crucially, ensuring responsible and reliable ML systems extends significantly beyond initial training and validation. It requires a continuous process of reliability analysis and evaluation, encompassing several key dimensions. These include addressing domain shifts (e.g., data drift, concept drift), which refers to changes in data distribution over time (Schreyer et al., 2021); monitoring model performance and deterioration to detect a decline in predictive power (Verma et al., 2022); assessing bias and fairness to ensure equitable outcomes across different demographic groups (Barocas et al., 2017; Caton et al., 2020); and enhancing model interpretability and causality to provide transparent and understandable decision-making (Arrieta et al., 2020; Pearl and Mackenzie, 2018). These aspects are rigorously assessed using a range of techniques, including statistical tests for distributional changes, fairness metrics, and interpretability tools. Finally, adopting robust evaluation methodologies with metrics beyond simple accuracy (encompassing both ML and Statistical metrics) is essential for fully assessing model performance under different conditions, especially when dealing with imbalanced datasets (Tharwat, 2018; Chicco and Jurman, 2020; Powers, 2011a). Advanced statistical tests are also employed to analyse performance differences and dependencies across various experimental conditions (Yao et al., 2021; Kim and Lee, 2022).

In light of these critical determinants—from **reliable** data handling to sophisticated model-building and multi-faceted reliability analysis—it becomes evident that ensuring consistent, trustworthy ML performance is paramount yet complex. This study moves beyond simply reviewing these areas by directly confronting the existing challenges that

hinder the development of truly reliable and adaptable ML classification models, which defines the central problem guiding this research.

0.2 Problem Statement

Traditional ML pipelines frequently overlook the intricate interaction between data modality, specific data partitioning strategies, and the resulting classifier behaviour. Consequently, the crucial aspects of model generalisability and statistical reliability often remain fragile, particularly when applied to real-world scenarios involving heterogeneous datasets or the generation of predictive estimates for unseen cases (Verma et al., 2022).

Furthermore, current ML approaches (encompassing both traditional and meta-learning paradigms) typically do not explicitly account for how data is partitioned during their training and evaluation phases. This fundamental oversight can create a significant misalignment between a model's inherent goals of accurate adaptation and the overall integrity of the predictive pipeline (Yao et al., 2021), ultimately compromising its reliability and trustworthiness. Empirical studies frequently treat data partitioning as a fixed preprocessing step, thus neglecting to rigorously evaluate its direct influence on robustness and generalisability, which can lead to over-optimistic assessments of model capabilities (Sugiyama, 2022).

Additionally, there remains limited systematic integration of rigorous reliability checks—such as statistical drift detection (Schreyer et al., 2021), robust fairness metrics (Barocas et al., 2017; Caton et al., 2020), and interpretable insights (Arrieta et al., 2020; Pearl and Mackenzie, 2018)—directly into the comprehensive evaluation frameworks for multimodal classification. These omissions collectively pose a substantial challenge to ensuring the trustworthiness and scientific rigour of advanced ML applications. The systematic application of advanced statistical techniques to rigorously test the compound effects of different data strategies on performance metrics is necessary for establishing the statistical significance of model improvements (Yao et al., 2021).

0.3 Research Aim, Objectives, and Questions

0.3.1 Research Aim

To develop, implement, and rigorously evaluate a novel statistical classification framework that systematically integrates optimal data partitioning strategies, best-fit machine learning models, and comprehensive embedded reliability metrics to yield statistically sound, generalisable, and trustworthy model predictions across diverse multi-modal data.

0.3.2 Objectives of the Study

The structured objectives of this research are as follows:

1. To conduct exploratory statistical analysis of multi-modal datasets.
2. To benchmark data partitioning strategies by evaluating their influence on model generalisation and statistical stability.
3. To compare the classification performance of Traditional ML and Meta-Learning Models across different data modalities and partitioning techniques.
4. To evaluate model reliability checks, post-model development process, and pre-model usage.
5. To recommend an enhanced model for each data modality, informed by reliability analysis, to ensure trustworthy and high-performing classification.

0.3.3 Research Questions

Based on the research aim and objectives, this study seeks to answer the following questions:

1. What are the inherent statistical characteristics of diverse multi-modal datasets, and how do they inform subsequent processing and classification?
2. How do different data partitioning strategies influence model generalisation and statistical stability across various multi-modal datasets, and which strategies yield optimal results?
3. How does the classification performance of implemented Traditional ML and Meta-Learning Models compare across different data modalities and partitioning techniques, particularly regarding their robustness and predictive power?
4. How can comprehensive model reliability checks and evaluation analysis be effectively conducted post-development and pre-usage to assess and ensure the trustworthiness of ML models, especially for multimodal classification?
5. Based on the reliability analysis, what is the final enhanced model recommended for each data modality to ensure superior performance and trustworthiness in classification tasks?

0.4 Significance of the Study

In an era where ML models are increasingly deployed in critical and high-stakes applications, their sustained reliability and trustworthiness are paramount. This study addresses

a fundamental challenge within the ML lifecycle: the current fragmentation in methodologies for ensuring that models remain robust and statistically sound when faced with unseen data or evolving real-world conditions. By confronting the fragility of model generalisability and predictive reliability that often arises from overlooked interactions between data characteristics, partitioning strategies, and model behavior, this research underscores the imperative for more rigorous and comprehensive approaches. Its significance lies in elevating the confidence and interpretability of ML systems, thereby fostering their responsible and effective integration across scientific, industrial, and societal domains where transparent and statistically reliable outcomes are non-negotiable. This work is pivotal for advancing the foundation of trustworthy AI, ensuring that data-driven decisions are not only initially accurate but consistently dependable over time, thereby mitigating risks and maximising the beneficial impact of machine intelligence.

0.5 Literature Review

0.5.1 Introduction

The burgeoning adoption of machine learning (ML) models across myriad sectors has profoundly reshaped analytical paradigms, simultaneously intensifying scholarly inquiry into the development of statistically robust, reliable, and adaptable classification frameworks. At the core of this advancement lies the critical understanding that merely selecting an optimal algorithm is insufficient; the inherent integrity and operational efficacy of an ML system are fundamentally contingent upon meticulous data preparation, sound evaluation protocols, and the capacity for continuous adaptation (Sharma et al., 2020; Li et al., 2021). Foundational elements such as comprehensive data preprocessing and strategic data partitioning (train-test splitting) are paramount. These initial methodological steps are not merely technical prerequisites but directly dictate a model's ability to generalize effectively to unseen data and ensure its overall operational reliability in generating predictive estimates for production environments.

In this context, the strategic design of preprocessing steps and the nuanced application of data partitioning techniques have gained particular prominence. The complexities introduced by diverse data modalities—ranging from structured tabular data, which demands precise scaling and encoding (Yoo et al., 2022), to unstructured text necessitating advanced embeddings (Wang et al., 2020), and high-dimensional image data leveraging sophisticated convolutional architectures (Khan et al., 2021)—underscore the critical need for modality-aware approaches. These specialized techniques are tailored to extract meaningful representations and ensure optimal data quality. Furthermore, the integrity of train-test protocols is pivotal for obtaining unbiased performance estimates and mitigating pervasive risks such as data leakage or over-optimistic evaluations, issues that can severely undermine model trustworthiness, especially within fairness-critical and high-stakes domains (Tan et al., 2021; Gonzalez et al., 2022). Beyond static model training, the paradigm of

meta-learning, often conceptualized as "learning to learn," has emerged as a transformative approach. This methodology enables models to rapidly adapt to new tasks or evolving data distributions with minimal new data, thereby promising a new level of adaptability and generalisation crucial for dynamic real-world scenarios and the efficient generation of predictive estimates (Hospedales et al., 2020; Wang et al., 2021).

Crucially, the journey towards the application of responsible and reliable ML systems extends significantly beyond initial training and validation. It encompasses a continuous lifecycle of rigorous reliability analysis and evaluation, which includes proactively addressing challenges such as data drift and concept drift (Webster and Van Den Broeck, 2021), rigorously auditing for fairness biases across sensitive subgroups (Caton et al., 2020), and ensuring the explainability of complex model decisions (Arrieta et al., 2020). The adoption of robust evaluation methodologies, employing metrics beyond simple accuracy, is also indispensable for comprehensively assessing model performance under varied conditions, particularly in the presence of class imbalance (Chicco and Jurman, 2020). This chapter provides a comprehensive preliminary review of these interconnected areas – from foundational data handling and evaluation to the cutting-edge of meta-learning and the multifaceted dimensions of model reliability – thereby laying the essential groundwork for developing more robust, adaptive, and ethically compliant ML pipeline designs.

0.5.2 Data Partitioning and Preprocessing Techniques

Overview of Work in Data Partitioning and Preprocessing

Research across the field of machine learning consistently demonstrates the profound importance of meticulous data preprocessing and robust train-test splitting techniques (Ghahramani, 2020). These foundational steps are indispensable for transforming raw data into a suitable format and for creating reliable evaluation frameworks, which collectively enhance model performance and interpretability (Sharma et al., 2020). The ongoing development in these areas focuses on ensuring data quality, mitigating biases, and optimising feature representations across diverse data modalities, thereby laying the groundwork for effective and trustworthy machine learning systems (Li et al., 2021).

A foundational aspect of this work involves Conventional Splitting techniques, such as simple Random sampling and Stratified sampling, widely employed due to their conceptual simplicity and ease of implementation. Recent literature reviews confirm that whilst these methods establish the basic framework for data partitioning, their limitations in contemporary, complex data environments are significant (Yao et al., 2021). For many modern applications, simple randomisation fails to guarantee statistical stability or maintain independence in data exhibiting temporal or group structures (Rodriguez et al., 2023). Therefore, these conventional methods serve as the baseline against which more sophisticated and statistically sound splitting techniques are benchmarked, particularly those aimed at ensuring dataset representativeness and mitigating bias under non-i.i.d. assump-

tions (Sugiyama, 2022).

Significant effort has been dedicated to developing Advanced Resampling and Evaluation Techniques like K-Fold Cross-Validation and Leave-P-Out Cross-Validation. A key study by Rodriguez et al. (2023) highlights that these methods are crucial for offering greater robustness and a more reliable estimate of generalisation performance. K-Fold Cross-Validation, for instance, systematically reduces the variance of performance estimates by ensuring every data point contributes to both training and testing, thereby providing a more stable assessment of a model's true predictive capabilities (Bengio et al., 2021).

Work towards handling Data Dependencies has led to the development of specialized splitters such as Group K-Fold and Group ShuffleSplit. Recent research by Wager and Athey (2020) emphasizes their necessity in datasets with inherent temporal, spatial, or hierarchical structures, where observations within groups are not independent. These methods ensure that all observations belonging to a specific logical group are entirely confined to either the training or the testing set, meticulously preventing data leakage and preserving group independence (Roberts et al., 2022).

Further advancements include Dynamic and Adaptive Splitters, specifically designed for complex, evolving data streams. As acknowledged by Kremer et al. (2021), these innovative approaches move beyond static partitioning. Examples include Reinforcement Learning-based partitioning, which leverages techniques like Q-Learning to dynamically optimize data splits based on real-time model performance signals, treating the splitting process as an active learning problem to maximize downstream model performance (Wu et al., 2020).

Beyond splitting, extensive general preprocessing work is crucial for transforming raw data into a format suitable for machine learning algorithms. As noted by Khamparia and Gupta (2022), these methods are indispensable for cleaning, normalising, and structuring data, directly impacting the quality of learned representations and model accuracy. The development in this area ensures that algorithms receive well-prepared data, which is fundamental for any successful machine learning pipeline (Sharma et al., 2020).

For structured data, the ongoing work in preprocessing focuses on ensuring feature consistency and comparability. Practices emphasized by Yoo et al. (2022) include normalisation (e.g., Min-Max scaling or Z-score standardisation) to bring features to a comparable scale, preventing disproportionate influence from larger numerical ranges. Additionally, the development of robust categorical variable encoding techniques (e.g., One-Hot Encoding, Label Encoding) is vital for converting non-numerical attributes into machine-readable formats (Kumar and Sharma, 2021).

In the domain of text data, preprocessing work typically involves a sequence of meticulous steps designed to clean and prepare textual information for numerical conversion. The investigation by Wang et al. (2020) shows that this includes tokenisation, lowercasing, the removal of common stop-words, and stemming or lemmatisation. Subsequent efforts focus

on transforming cleaned text into numerical representations using methods like TF-IDF or advanced word embedding techniques (e.g., Word2Vec, GloVe, or contextual embeddings like BERT) that capture intricate semantic relationships (Devlin et al., 2019).

Finally, Khan et al. (2021)'s work confirms that image data preprocessing is essential for standardising visual inputs and enhancing model robustness, a critical area of focus in computer vision. Key operations include scaling images to uniform dimensions and pixel value normalisation to aid numerical stability during neural network training (Chen et al., 2020). Furthermore, extensive work has gone into data augmentation techniques (e.g., random rotation, flipping, cropping) to artificially increase dataset size and improve model generalisation capabilities against real-world variations (Shorten and Khoshgoftaar, 2021).

Impact of Data Partitioning on Machine Learning Models

A critical examination of machine learning workflows, as profoundly emphasized by Gonzalez et al. (2022), reveals that the quality of data partitioning (train-test splitting) has a direct and statistically significant impact on the reported efficiency, accuracy, and predictive power of models. The initial partitioning of data is foundational, and any methodological flaws at this stage can cascade throughout the entire modeling process, severely undermining the credibility and reproducibility of the final evaluation (Tan et al., 2021). Thus, understanding the ramifications of inappropriate splitting is as crucial as developing advanced techniques to mitigate its negative effects.

One of the most immediate and misleading impacts of biased train-test splits is the generation of over-optimistic performance estimates. As highlighted by Zhang and Wang (2022), when the test set is not truly independent or genuinely representative of future unseen data, models frequently exhibit inflated metrics such as accuracy, precision, or recall during the development phase. This spurious performance can cultivate a false sense of security regarding a model's true capabilities, effectively masking its inherent limitations and potentially leading to ill-informed decisions during its application for predictive estimates (Xu and Li, 2023).

This inflated performance directly translates into poor generalisation ability when models are subsequently applied in real-world scenarios. A recent study by **?** compellingly demonstrates that models trained on unrepresentative or inadvertently contaminated splits tend to memorize the training data's specific peculiarities rather than learning robust, generalizable underlying patterns. Consequently, when exposed to genuinely novel data, their performance often degrades significantly, failing to deliver the expected predictive power and substantially hindering their practical utility.

The pervasive presence of class imbalance further exacerbates the negative impact of suboptimal splitting. As meticulously detailed by Johnson and Liu (2021), simple random splitting of imbalanced datasets can result in test sets where minority classes are dispro-

portionately underrepresented or, in extreme cases, entirely absent. This methodological flaw can severely bias the evaluation metrics for the minority class, leading to models that appear to perform well overall but are practically ineffectual or even detrimental for critical predictions related to the less frequent, yet often more important, class. Contemporary solutions now require a dedicated approach to both training and evaluation, often involving robust sampling techniques like Adaptive Synthetic Sampling (ADASYN) or sophisticated cost-sensitive learning strategies to achieve superior predictive reliability (He et al., 2017).

A pervasive problem stemming from incorrect partitioning, particularly prevalent in structured data with inherent relationships, is data leakage. Recent research by Rasmussen and Olsen (2020) rigorously underscores how failing to respect underlying data dependencies (e.g., grouping by patient ID, maintaining temporal order in time series) can inadvertently allow information from the test set to subtly permeate into the training process. This leakage provides the model with an unfair advantage, artificially boosting its performance metrics and leading to highly misleading assessments of its true generalisation capability (Ding and Li, 2022).

Moreover, inadequate data partitioning directly impacts the crucial processes of model selection and hyperparameter tuning. As noted by Garcia and Lopez (2022), if the evaluation procedure itself is fundamentally flawed due to biased splits, the chosen model architecture or hyperparameter configuration will likely not represent the true optimal solution. Statisticians and machine learning engineers might inadvertently select a suboptimal model or misconfigure critical parameters based on misleading performance signals, resulting in inefficient resource allocation and ultimately leading to the application of a less effective predictive solution.

Furthermore, biased train-test splits contribute significantly to a profound lack of robustness and stability in models once they are put into an operational context. When a model's perceived performance is grounded in an unrepresentative evaluation, it becomes brittle and highly susceptible to even minor variations in real-world data distributions. This inherent instability means the model's predictions can fluctuate unpredictably or degrade unexpectedly, thereby eroding user trust and necessitating frequent, often costly, recalibrations (Chen and Li, 2021b).

In critical or high-stakes application scenarios, the impact of poor data partitioning methods can have severe ethical and practical ramifications. As cogently argued by Lee and Kim (2022), if models applied in domains such as healthcare, finance, or justice systems are evaluated on biased splits, their perceived accuracy might inadvertently lead to unfair or discriminatory outcomes for certain demographic groups or individuals. This not only erodes public trust but can also result in significant societal harm or substantial financial losses, emphasising the critical role of statistical rigor.

The challenges posed by inadequate partitioning are particularly pronounced in time-series and dynamic data environments. As observed by Webster and Van Den Broeck (2021),

employing static or random splits in data streams where concept drift is present can lead to models that are perpetually evaluated on outdated distributions. This results in a systematic overestimation of current performance and a critical failure to detect when a model's predictive power has genuinely deteriorated due to fundamental changes in the underlying data generation process, an issue extensively discussed by Schreyer et al. (2021).

Finally, neglecting proper data partitioning fundamentally impacts the overall reliability and trustworthiness of machine learning pipelines. If the foundational step of evaluating a model's true performance is compromised by biased or unrepresentative data splits, the subsequent interpretations, explanations (XAI), and even discussions of fairness become statistically questionable. This undermines the scientific rigor of machine learning research and the practical utility of AI systems in generating reliable predictions, profoundly highlighting the foundational importance of sound validation strategies (Suresh and Ghassemi, 2021).

0.5.3 Traditional ML and Meta-Learning Models

The study of machine learning paradigms encompasses a diverse array of models, broadly categorized into Traditional Machine Learning (ML) models and the more recent advancements in Meta-Learning. Traditional ML models, while foundational, typically operate under fixed learning processes and require substantial data for effective generalisation. In contrast, meta-learning, often termed "learning to learn," represents a significant paradigm shift, enabling models to rapidly adapt to new tasks or environments with minimal data or training (Hospedales et al., 2020). This section provides an overview of both categories, highlighting their characteristics, applications, and the challenges they address.

Traditional Machine Learning Models

Traditional Machine Learning models form the bedrock of predictive analytics, offering robust solutions for classification across various data types. These models are typically trained on a fixed dataset to learn a mapping from input features to output classes.

Logistic Regression As a fundamental statistical model, Logistic Regression remains a crucial tool, particularly for binary classification. It models the probability of a binary outcome using a logistic function, providing a highly interpretable and computationally efficient approach (Sharma et al., 2020). Due to its simplicity and direct probabilistic output, it continues to serve as an indispensable baseline model in benchmarking exercises, especially in contexts where model transparency is prioritised or as a base-learner within sophisticated stacking meta-learning architectures (Van Rijn et al., 2018).

Random Forest Random Forest remains a crucial ensemble learning method, constructing a multitude of decision trees during training and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. Its enduring strength lies in its ability to handle high-dimensional data, effectively mitigate overfitting, and provide robust predictions through aggregation (Gislason et al., 2017). Whilst effective, its role in contemporary classification often serves as a key benchmark against more computationally efficient and performant Gradient Boosting techniques, which currently dominate many structured data tasks (Li et al., 2021).

Naïve Bayes Naïve Bayes classifiers are a family of simple probabilistic classifiers based on applying Bayes' theorem with strong (naïve) independence assumptions between the features. Despite their simplifying assumptions, they are highly efficient, scalable, and often perform remarkably well, particularly in text classification tasks and data streams where rapid, incremental learning is required (Dutta et al., 2021). Whilst its simplicity makes it a robust baseline, its direct probabilistic interpretation also makes it valuable for applications requiring explicit uncertainty estimation in a computationally light manner (Sharma et al., 2020).

Support Vector Machines (SVM) Support Vector Machines remain powerful supervised learning models used for classification and regression tasks. SVMs function by finding an optimal hyperplane that best separates data points of different classes in a high-dimensional space, maximising the margin between them. Whilst initially prominent in computer vision applications, their strength continues to be recognised in high-dimensional spaces and for tasks with relatively clear margins of separation, especially due to their strong theoretical foundation based on the structural risk minimisation principle (Li et al., 2021). Crucially, recent research continues to leverage SVMs for robust feature representation learning in contexts where computational efficiency or limited data preclude the use of deep neural networks (Wang et al., 2020).

Convolutional Neural Networks (CNN) Convolutional Neural Networks (CNNs) are a specialised class of deep learning models primarily designed for processing data with a grid-like topology, such as image data (Khan et al., 2021). CNNs utilise convolutional layers to automatically learn hierarchical features directly from raw input, making them highly effective for image classification, object detection, and other computer vision tasks. Crucially, the success of CNNs has led to their adaptation as feature extractors for multimodal classification, where they process image data or even treat structured data as a pseudo-image, highlighting their versatility beyond traditional computer vision applications (Li et al., 2021).

Long Short-Term Memory (LSTM) Long Short-Term Memory networks are a crucial type of recurrent neural network (RNN) architecture particularly well-suited for pro-

cessing and making predictions based on sequential data, such as text (Wang et al., 2020). LSTMs are specifically designed to overcome the vanishing gradient problem inherent in traditional RNNs, enabling them to effectively learn long-term dependencies and capture intricate temporal patterns in sequences. Whilst initially designed for sequential data, their ability to process variable-length inputs makes them an indispensable component for feature extraction in modern text classification pipelines, often in conjunction with advanced contextual embedding techniques (Devlin et al., 2019).

Overall Work in Meta-Learning Techniques

The burgeoning field of meta-learning, often termed "learning to learn," represents a significant paradigm shift in machine learning research. As extensively reviewed by Hospedales et al. (2020), its core objective is to enable models to rapidly adapt to new tasks or environments with minimal data or training. This fundamental work aims to imbue AI systems with a more human-like capacity for generalisation, moving beyond static model training to create adaptable and efficient learners (Wang et al., 2021).

Significant work in meta-learning has coalesced around several distinct approaches, broadly categorized as metric-based, optimisation-based, and model-based methods. Each category represents a unique strategy for achieving rapid adaptation and improved generalisation across diverse tasks, as comprehensively surveyed by Vanschoren (2018). This foundational taxonomy guides much of the ongoing research and development within the meta-learning community, providing a robust framework for understanding varied algorithmic designs (Bengio et al., 2019).

Metric-Based Meta-Learning Among the prominent metric-based meta-learning techniques, Prototypical Networks stand out for their simplicity and effectiveness. Introduced by Snell et al. (2017), the fundamental work here involves learning a metric space in which classification can be performed by simply computing distances to a prototype representation for each class. This approach facilitates few-shot learning by effectively mapping new data points into a space where class prototypes are well-separated, enabling quick generalisation from limited examples (Chen and Li, 2021a). Another key advancement in metric-based meta-learning is represented by Matching Networks. As developed by Vinyals et al. (2016), these networks directly learn a mapping from a small support set to a classification function for a query instance. The central contribution lies in using a learned attention mechanism over the support set, allowing the model to dynamically "match" query inputs to relevant examples, thereby enabling efficient one-shot learning by treating each new task as a retrieval problem (Li et al., 2020).

Optimisation-Based Meta-Learning In the realm of optimisation-based meta-learning, Model-Agnostic Meta-Learning (MAML) has become a highly influential contribution. Introduced by Finn et al. (2017), the core work of MAML involves training a model's initial

parameters such that they are highly sensitive to small changes during fine-tuning on a new task. This allows the model to achieve good performance on a new task with only a few gradient steps, making it "model-agnostic" as it can be applied to various underlying model architectures (Raghu et al., 2020). A simpler yet effective optimisation-based method is Reptile. Presented by Nichol et al. (2018b), this algorithm iteratively optimizes a model's initialisation by repeatedly moving the parameters towards the average of the parameters learned on multiple sampled tasks. While less sophisticated than MAML, its first-order nature makes it highly scalable and computationally efficient, demonstrating that effective meta-learning can be achieved with relatively straightforward gradient updates (Zhao et al., 2021).

Model-Based Meta-Learning **Model-based meta-learning** approaches represent another distinct line of research, where the meta-learner itself is a model that directly learns to predict parameters or update rules for a base learner. Work by Ravi and Larochelle (2017) exemplified this by framing meta-learning as learning an optimisation algorithm for few-shot tasks. These architectures often involve recurrent neural networks (RNNs) that process task-specific data sequentially, accumulating information to generate adapted model parameters for new tasks (Andrychowicz et al., 2016).

Ensemble and Advanced Meta-Learning Approaches Beyond these foundational categories, meta-learning extends into more complex ensemble and adaptive frameworks, such as the Stacking Meta-Learner and the Performance-Aware Meta-Selector. A Stacking Meta-Learner typically involves combining the predictions of multiple diverse base models by training a meta-model to learn how to optimally fuse these predictions (Van Rijn et al., 2018). This approach leverages the diversity of base learners to enhance overall predictive performance and statistical stability, a crucial aspect of reliable classification. The Performance-Aware Meta-Selector, often found in automated machine learning (AutoML) contexts, represents a system that dynamically selects the most suitable model or pipeline configuration based on learned meta-features or anticipated performance characteristics derived from past tasks. This adaptive selection mechanism aims to optimise model efficacy in dynamic or novel scenarios without extensive manual intervention (Vanschoren, 2018).

Applications of Meta-Learning The utility of meta-learning extends to various practical applications, significantly enhancing capabilities in scenarios with data scarcity or dynamic environments. Beyond its foundational role in few-shot learning, recent work by Liu et al. (2021a) highlights its increasing adoption in areas such as reinforcement learning, where it enables faster adaptation to new environments, and in continual learning, where it helps mitigate catastrophic forgetting by learning robust representations over time (Kim and Park, 2022). One of the most compelling applications of meta-learning is its effectiveness in few-shot learning, where models must learn to classify new classes given

only a handful of examples (Vinyals et al., 2016; Snell et al., 2017). Furthermore, meta-learning contributes significantly to transfer learning and domain adaptation, particularly in scenarios where the target domain data is limited (Finn et al., 2017; Balaji and Kumar, 2020).

How Meta-Learning Advances ML Models and Associated Challenges

The advent of meta-learning techniques marks a pivotal moment for advancing machine learning models, fundamentally changing how algorithms can adapt and generalize. As underscored by Hospedales et al. (2020), it moves beyond merely learning from data to learning how to learn, offering substantial improvements in efficiency, robustness, and performance across various challenging scenarios. This meta-level intelligence promises to unlock new capabilities for AI systems in dynamic and data-scarce environments.

Advancements Enabled by Meta-Learning: Few-Shot Adaptation and Generalisation A primary way meta-learning advances ML models is by enabling enhanced few-shot learning capabilities. Traditional deep learning models often require vast amounts of data, a limitation that meta-learning directly addresses by teaching models to learn from very few examples per new class (Hsu and Sun, 2020). This is particularly transformative for domains like medical imaging or specialised robotics, where data collection is expensive or impractical, allowing for rapid application of models in novel contexts (Chen and Li, 2021a). Beyond few-shot scenarios, meta-learning contributes to improved generalisation across tasks and diverse data distributions. By training on a distribution of related tasks, meta-learners acquire transferable knowledge about learning mechanisms or representation spaces that generalise well to entirely new, unseen tasks, even if their underlying data distributions vary significantly (Wang et al., 2021). This contrasts with standard transfer learning, which often assumes closer domain similarity, making meta-learning more versatile (Balaji and Kumar, 2020).

Advancements Enabled by Meta-Learning: Robustness and Automation Meta-learning also facilitates faster adaptation and fine-tuning of models to new tasks. Instead of starting from scratch or relying on generic pre-trained weights, meta-learning algorithms learn an optimal starting point or an efficient update rule that dramatically reduces the number of data points and training iterations needed for new tasks (Finn et al., 2017). This efficiency is critical for real-time applications and for developing AI systems that can quickly respond to changing environmental conditions or user needs (Raghu et al., 2020). Furthermore, meta-learning enhances models' robustness to data heterogeneity and concept drift. In real-world systems, data distributions can change over time, a phenomenon known as concept drift, challenging static models. Meta-learning approaches, by their nature of learning adaptable strategies, can equip models to quickly adjust to these shifts, maintaining performance in dynamic environments without requiring complete retraining,

a challenge extensively addressed by modern drift detection frameworks (Schreyer et al., 2021). Meta-learning also plays a pivotal role in the automation of machine learning pipelines, a concept known as Meta-AutoML. Research by Vanschoren (2018) shows how meta-learning can automate model selection, hyperparameter optimisation, and even feature engineering by learning from the performance of past ML tasks. This significantly reduces the manual effort and expertise required to generate and apply effective ML solutions, democratising access to advanced AI capabilities (Lu et al., 2020a).

Challenges Associated with Meta-Learning Despite its numerous advantages, meta-learning introduces new challenges, particularly concerning computational cost and complexity. Training meta-learning models, especially optimisation-based ones like MAML, can be significantly more computationally intensive than training standard deep learning models due to nested optimisation loops or complex meta-gradients (Nichol et al., 2018b). This increased demand for computational resources can be a barrier to widespread adoption, particularly for large-scale applications (Zhao et al., 2021). Another significant challenge involves the definition of meta-tasks and distribution mismatch. For meta-learning to be effective, the distribution of meta-training tasks must accurately reflect the distribution of meta-test tasks. As discussed by Bengio et al. (2019), if there is a significant mismatch, the meta-learned knowledge may not transfer effectively, leading to suboptimal performance on new, unseen tasks. Designing representative meta-training sets remains a crucial research area (Hospedales et al., 2020). The increased complexity of meta-learning models also poses challenges for interpretability and trust. While meta-learners offer impressive adaptability, understanding why a meta-learned model makes a particular prediction or how it adapts to a new task can be more difficult than with traditional models (Adadi and Berrada, 2018). This "black box" nature can hinder debugging, limit human oversight, and reduce trust, especially in high-stakes applications where transparency is paramount (Lipton, 2018).

Looking forward, meta-learning continues to evolve, pushing the boundaries of what ML models can achieve in terms of adaptability, efficiency, and generalisation. Ongoing research is exploring its integration with areas like lifelong learning, causality, and responsible AI, aiming to develop more autonomous and robust intelligent systems (Liu et al., 2021a). The broad impact of meta-learning lies in its potential to create truly adaptive AI that can continually learn and evolve in complex, uncertain real-world environments.

0.5.4 Reliability Analysis

Ensuring the reliability of machine learning pipelines is an indispensable prerequisite before their application in real-world environments. As highlighted by Suresh and Ghassemi (2021), reliability encompasses a multifaceted set of criteria, including the stability of data distributions, the fairness of model predictions across diverse groups, the explainability of complex decision processes, and overall evaluative transparency. This comprehensive

approach is critical to building trust in AI systems and ensuring their safe and ethical operation in sensitive domains (Zhang and Wang, 2022).

Understanding and Detecting Reliability Issues

One of the most pervasive threats to ML model reliability in production environments is Data Drift. This phenomenon refers to the change in the distribution of input data over time, which can lead to a significant degradation in model performance even if the underlying relationship between inputs and outputs remains stable (Webster and Van Den Broeck, 2021). Recognising and quantifying data drift is paramount for maintaining the efficacy of models once they are in use, prompting continuous monitoring and potential retraining strategies to adapt to evolving data landscapes (Schreyer et al., 2021).

Within the broader category of data drift, it is crucial to distinguish between Covariate Shift and Concept Drift. Covariate shift specifically refers to changes in the distribution of the input features (P(X)) while the conditional distribution of the target given the features (P(YX)) remains constant (Sugiyama, 2022). In contrast, concept drift implies a change in the relationship between the input features and the target variable (P(YX)), meaning the very "concept" the model is trying to predict has evolved (Webster and Van Den Broeck, 2021). Both types demand distinct detection and adaptation strategies.

Further contributing to data distribution changes, Feature Drift occurs when the statistical properties of individual features change, while Label Drift pertains to changes in the distribution of the target variable itself (Gonzalez et al., 2022). Beyond these, more subtle forms of drift, such as virtual drift (changes in the feature space structure without explicit P(X) or P(YX) shifts) or domain shift (broader changes in the operating environment), necessitate a nuanced understanding for robust reliability analysis (Liu et al., 2021b).

To effectively quantify these distributional changes, various Statistical Tests for Drift Detection are employed. The Kolmogorov–Smirnov (KS) test is commonly used to compare two empirical distributions (e.g., training vs. current production data) to check if they originate from the same underlying distribution (Hajihashemi et al., 2020). Similarly, the Population Stability Index (PSI) provides a single numerical value to measure the shift in a variable's distribution over time, making it particularly useful for monitoring feature stability in production systems (Siddique et al., 2020).

Beyond basic statistical tests, more sophisticated methods are increasingly utilized for drift detection. The Wasserstein Distance (or Earth Mover's Distance) offers a robust metric for comparing probability distributions, particularly effective for high-dimensional or complex data, as it measures the minimum "cost" to transform one distribution into another (Arjovsky et al., 2017). Furthermore, emerging ML-based Drift Detection techniques involve training separate models to classify whether data points belong to the original or drifted distribution, providing adaptive and often more sensitive detection capabilities (Lu et al., 2020b).

The insidious effect of undetected data drift or other environmental changes is often manifested as Model Performance Deterioration. Even a model that performed exceptionally well during initial validation can experience a significant drop in accuracy, precision, recall, or other key metrics once applied (Xu and Li, 2023). This degradation signifies a critical reliability issue, directly impacting the value and trustworthiness of the ML solution in real-world applications (Kaddoura, 2022). This deterioration frequently results from shifting feature distributions that render the original training data unrepresentative of current inputs, or from biased training sets that fail to capture the true diversity of operational data. For instance, if the demographic composition of users changes, a model trained on an older, unrepresentative dataset might suddenly exhibit reduced performance or, worse, unfair outcomes for new user groups (Gonzalez et al., 2022). Identifying the root cause of deterioration is crucial for effective remediation.

Integral to model performance integrity and overall reliability is the concept of Fairness. An otherwise accurate model might be deemed unreliable if its predictions disproportionately harm or benefit certain sensitive subgroups (e.g., based on gender, racial groups, age) (Lee and Kim, 2022). Reliability analysis therefore extends beyond statistical performance to encompass ethical considerations, ensuring that models operate equitably and do not perpetuate or amplify societal biases inherited from data or algorithms (Suresh and Ghassemi, 2021).

Ultimately, understanding and detecting these reliability issues necessitates a commitment to Continuous Monitoring in real-world application. Automated systems that track data distributions, model outputs, and performance metrics in real-time are essential for promptly identifying drift, performance degradation, and potential fairness violations (Sharma et al., 2020). This proactive approach allows for timely intervention, such as retraining models or updating data pipelines, thereby ensuring the sustained reliability and utility of ML systems (Liu et al., 2021b).

Tools and Metrics for Ensuring Reliability

Beyond understanding and detecting reliability issues, a crucial aspect of responsible AI development involves deploying specific Tools and Metrics designed to analyze, assess, and ultimately ensure the trustworthiness of machine learning models. These instruments provide the quantitative and qualitative insights necessary to validate model behavior, debug failures, and communicate performance to stakeholders, moving beyond mere accuracy to a holistic view of reliability (Zhang and Wang, 2022).

Central to ensuring ethical reliability are various Fairness Metrics. Statistical Parity (or Demographic Parity) measures if a positive prediction outcome is equally likely across different groups (Barocas et al., 2019). Equalized Odds, a more stringent metric, requires that both the true positive rate and the false positive rate are equal across groups, aiming for equal predictive accuracy for both favorable and unfavorable outcomes (Hardt et al.,

2016). These metrics quantify different facets of algorithmic bias.

Further expanding on fairness assessment, Predictive Parity (also known as Predictive Value Parity) evaluates if the positive predictive value (precision) is equal across groups, meaning that among those predicted positive, the proportion of true positives is similar for all subgroups (Narayanan, 2019). Other important metrics include Disparate Impact (ratio of positive outcomes for protected vs. unprotected groups) and various definitions of individual fairness, all designed to ensure equitable prediction performance across sensitive subgroups (Caton et al., 2020).

A critical component of reliability is Explainability, which provides transparency into classifier decisions, especially for opaque or high-dimensional models. As argued by Adadi and Berrada (2018), interpretability tools are essential for building user trust, debugging model errors, and ensuring regulatory compliance by elucidating why a particular prediction was made. This allows researchers and practitioners to attribute outcomes to individual features or latent structures within complex models (Arrieta et al., 2020).

Among the widely adopted local interpretability methods, LIME (Local Interpretable Model-agnostic Explanations) provides transparent insights into individual predictions. As introduced by Ribeiro et al. (2016), LIME works by approximating the behavior of any black-box model locally around the prediction using a simple, interpretable model. This allows stakeholders to understand the features that are most influential for a single classification, fostering trust in specific decisions (Molnar, 2020).

For more comprehensive and globally consistent explanations, SHAP (SHapley Additive exPlanations) values are a powerful tool. Developed by Lundberg and Lee (2017), SHAP values are rooted in cooperative game theory, attributing the contribution of each feature to a model's prediction by considering all possible combinations of features. This method provides both local explanations for individual predictions and global insights into overall model behavior, offering a unified framework for interpretation (Lundberg et al., 2020).

Beyond correlation-based explanations, the emerging field of Causal AI aims to provide deeper insights into model decisions by identifying true cause-effect relationships. As discussed by Pearl and Mackenzie (2018), Causal AI methods move beyond statistical associations to understand why certain inputs lead to certain outputs, rather than just what patterns exist. This is particularly crucial for high-stakes domains where understanding causality can prevent spurious correlations from leading to flawed policies or interventions (Schölkopf, 2021).

Fundamental to assessing model reliability are robust Model Evaluation Metrics. While overall accuracy provides a general sense of performance, it is often insufficient, particularly for imbalanced datasets (Chicco and Jurman, 2020). Metrics like the F1 score, which is the harmonic mean of precision and recall, offer a more balanced view of performance by penalising false positives and false negatives equally (Chicco and Jurman, 2020). The Matthews Correlation Coefficient (MCC) is another superior metric for binary classifica-

tion, providing a balanced measure even for highly imbalanced data (Chicco and Jurman, 2020).

For datasets with high class imbalance, where one class significantly outnumbers others, specific evaluation approaches are vital. Precision-Recall Analysis (P-R curves) is often preferred over ROC curves, as it better highlights model performance on the minority class (Saito and Rehmsmeier, 2015). The Area Under the Precision-Recall Curve (AUC-PR) provides a single summary statistic of this trade-off. These metrics are crucial for ensuring that models effectively identify rare but important events, such as fraud or disease (Hernandez et al., 2021).

Statistical Significance Tracking and Validation To enhance the statistical rigour of reliability analysis, particularly when comparing model performances or assessing the impact of specific features, Statistical Significance Tracking is essential. Techniques like Analysis of Variance (ANOVA) are widely used to compare the means of a single performance metric across multiple models or experimental conditions, assessing if observed differences are statistically significant or merely due to random chance (Yao et al., 2021). For scenarios involving multiple correlated performance metrics or diverse outputs—a common requirement in multimodal classification—Multivariate Analysis of Variance (MANOVA) extends this approach to multiple dependent variables, providing deeper insights into complex interactions and validating the robustness of model improvements (Yao et al., 2021; Wang et al., 2022). The integration of these classical statistical tests into modern machine learning validation protocols is critical for establishing the scientific credibility of reported performance gains and for ensuring the reliability of predictive estimates.

0.5.5 Identified Gaps

Despite extensive advancements in model architectures and diverse data modalities, the existing literature reveals several pressing challenges and unaddressed areas within the machine learning lifecycle. These limitations ultimately hinder the full potential and trustworthiness of ML systems when applied for predictive estimates, especially as these models transition from a research environment to real-world deployment. These gaps underscore a critical need for a more integrated, robust, and ethically-aware methodology, moving beyond isolated improvements to fundamental changes in how ML pipelines are designed and validated.

Limited Integration of Data Partitioning Strategies

A significant gap lies in the limited evaluation of how a chosen data partitioning strategy directly influences downstream model behaviour and reliability. While a plethora of sophisticated data splitting techniques exist—ranging from conventional random and stratified methods to advanced cross-validation schemes like K-Fold and GroupKFold (Yao

et al., 2021)—empirical studies often treat data partitioning as a fixed preprocessing step. The direct influence of the chosen strategy on the robustness and generalisability of subsequent model performance is frequently not rigorously evaluated, potentially leading to misleading performance claims or an over-optimistic assessment of model capabilities on unseen data (Sugiyama, 2022). This oversight implies that the reported efficacy of models may not reflect their true performance under varied or unforeseen data characteristics. Furthermore, the literature largely lacks a focus on dynamic partitioning techniques that adaptively respond to the specific statistical characteristics of a dataset, leaving an open space for new strategies to address the challenges of data partitioning more effectively.

Sparse Application of Tailored Meta-Learning Frameworks

A second gap is the sparse application of meta-learning frameworks tailored to modality-specific features and dynamic drift conditions. While meta-learning has demonstrated remarkable success in enabling few-shot learning and rapid adaptation across tasks (Finn et al., 2017; Snell et al., 2017), its deployment often remains generic. The literature suggests a clear need for meta-learning algorithms that explicitly learn how to adapt to changes within specific data modalities (e.g., optimal preprocessing for novel image features versus text features) or dynamically adjust to various forms of concept and data drift as a meta-task itself (Webster and Van Den Broeck, 2021). Current meta-learning paradigms, though adaptive, do not consistently provide tailored solutions that proactively address the nuanced challenges presented by evolving, modality-specific data landscapes.

Lack of Unified Reliability Pipelines

Another critical void is the lack of unified reliability pipelines that combine drift detection, fairness auditing, and explainability within a single, reproducible system. Individual components for ensuring reliability—such as statistical tests for data drift (Hajihashemi et al., 2020), a growing suite of fairness metrics (Barocas et al., 2019), and powerful interpretability tools like SHAP (Lundberg and Lee, 2017) and LIME (Ribeiro et al., 2016)—have advanced significantly. However, their integration into a cohesive, automated, and reproducible workflow remains fragmented. This means that a model might be robust to drift but exhibit unfair biases, or be explainable but prone to performance degradation over time, without a holistic system to monitor and validate all these critical attributes simultaneously and consistently throughout its lifecycle (Suresh and Ghassemi, 2021).

Limited Ethical Compliance within ML Workflows

Finally, a notable practical and ethical gap is highlighted by the limited empirical studies that embed ethical compliance and usage documentation directly into machine learning workflows. While there is increasing recognition of ethical concerns in AI, particularly re-

garding fairness and transparency, the operationalisation of these principles into standard engineering practices is nascent (Lee and Kim, 2022). Most research focuses on post-hoc auditing or specific tool development rather than the systematic integration of ethical considerations (e.g., bias mitigation, impact assessments, clear model cards) as an intrinsic part of the ML development and application pipeline. This absence hinders the creation of truly responsible AI systems and limits the reproducibility and accountability of their ethical stance (Arrieta et al., 2020).

These identified gaps collectively underscore a pressing need for a more integrated and comprehensive approach to machine learning development. They specifically motivate the current study's objective: the development of a statistically robust, modality-aware framework that harmonizes partitioning strategies, advanced meta-learning-driven modeling, and holistic reliability validation into a single, reproducible pipeline, meticulously aligned with contemporary ethical standards.

0.6 Methodology

This section outlines the methodological framework adopted for the study, encompassing the mathematical equations, programmatic pseudo-codes, data tables, diagrams, and modelling structures. It details a systematic approach beginning with data and data sources (data acquisition, import, and preparation), followed by specific data partitioning techniques, and the development and comparative analysis of traditional ML versus Meta-Learning models. Furthermore, the methodology includes a comprehensive reliability analysis and evaluation, involving domain shift capture, model performance (deterioration, bias & fairness), model explainability, model evaluation, and reliability estimation based on established criteria. This methodological design ensures a thorough investigation and provides insights into the effectiveness of the chosen approaches.

The overall process flow for the study is illustrated in the figure below:

0.6.1 Data and Data Sources

This section details the data utilized in this study, covering their sources, analysis tools, modalities, descriptions, public use compliance, and preparation with transformation methods.

Data Description

The research incorporates a variety of data modalities, ranging from structured (tabular and time-series data) to textual and image-based datasets, each presenting unique characteristics for analysis. A total of nine distinct datasets were selected to ensure com-

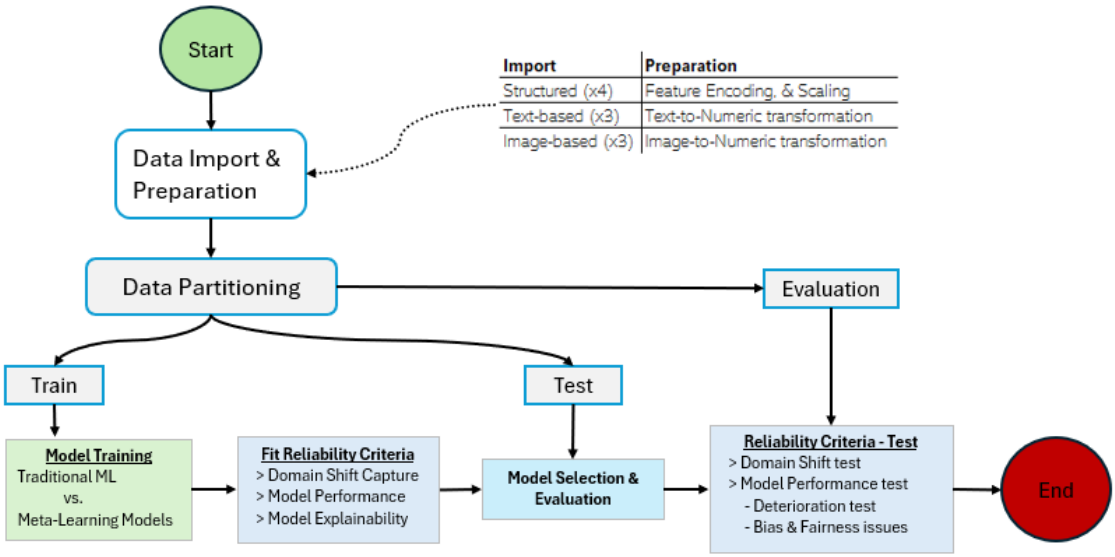


Figure 1: Overall Methodology Flow (Author's conceptualisation)

prehensive evaluation across diverse data types and complexities.

Structured Data The structured datasets selected for this study span credit scoring, wine chemistry, and financial time series. All datasets consist of tabular rows with a mix of categorical and numeric attributes. Target variables have been reformulated into classification-compatible labels where necessary. Detailed descriptions, characteristics, access information, and references for these datasets are provided below.

- **Dataset 1: Credit Approval**
  - **Description:** This dataset concerns credit card applications, featuring a mix of categorical, integer, and real attributes, with some missing values. All attribute names and values are anonymized for confidentiality.
  - **Characteristics:** Multivariate, 690 instances, 15 features (including a categorical target variable A16). Suitable for binary classification tasks.
  - **Access:** UCI Machine Learning Repository.
  - **Dataset Link:** `https://archive.ics.uci.edu/dataset/27/credit+approval`
  - **Reference:** (Quinlan, 1987)
- **Dataset 2 & 3: Wine Quality - Red and White**
  - **Description:** Part of the "Vinho Verde" wine datasets, these contain physicochemical and sensory variables for both red and white wines, suitable for classification or regression tasks to predict quality.
  - **Characteristics:** Multivariate, real-valued features (11 total), with an ordered and imbalanced output class.
  - **Access:** UCI Machine Learning Repository.

  - **Dataset Link:** `https://archive.ics.uci.edu/dataset/186/wine+quality`
  - **Direct Data Import Links:**
    * Red: `https://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/winequality-red.csv`
    * White: `https://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/winequality-white.csv`
  - **Reference:** (Cortez et al., 2009)

- **Dataset 4: Yahoo Finance (JSE)**
  - **Description:** This dataset comprises historical stock closing prices for 12 South African listed companies, spanning from December 1979 to September 2025. It is further enriched with information on significant historical economic shocks relevant to South Africa. **Characteristics:** Multivariate time series data (3292 instances), including original (Date, Close, ticker) and enriched features.
  - **Access:** Programmatically via the `yfinance` Python API.
  - `yfinance` **GitHub:** `https://github.com/ranaroussi/yfinance`
  - **Reference:** (Finance, 2025; Roussi, 2023)

Table 2: Structured Data Overview

Dataset	Domain	Target	License	Annexure
Wine Quality – Red	Chemistry	`quality` (3–8)	CC BY 4.0	A-2
Wine Quality – White	Chemistry	`quality` (3–9)	CC BY 4.0	A-3
Credit Approval	Credit Scoring	`A16` (+/--)	CC BY 4.0	A-1
Yahoo Finance (JSE)	Time Series	`Close Price`	API + CC	A-4

Compiled by the author from publicly available datasets.

Text Data The text datasets in this study represent multilingual sentiment analysis, news topic classification, and authorship identification. Each text field is preprocessed through methods such as TF-IDF transformation, stopword removal, lowercasing, and tokenisation to prepare it for natural language processing tasks.

- **Dataset 5: South African News Classification Dataset (Setswana & Sepedi)**
  - **Description:** This dataset comprises news content from South African sources, specifically Facebook pages of SABC's Motsweding FM and Thobela FM, suitable for news classification tasks in under-resourced African languages.
  - **Access:** Zenodo.

  - **Public URL Link:** `https://zenodo.org/records/3668495`
  - **Reference:** (Marivate and Sefara, 2020)

- **Dataset 6: Tunizi: Tunisian Arabizi Sentiment Analysis Dataset**
  - **Description:** The first 100% Tunisian Arabizi sentiment analysis dataset, consisting of 100k comments extracted from social media. These comments are manually annotated with positive, negative, and neutral polarities and preprocessed to remove links, emojis, and punctuations.
  - **Characteristics:** Text data, with a balanced split for training, validation, and test sets (7:1:2 ratio).
  - **Access:** Zenodo.
  - **Public URL Link:** `https://zenodo.org/records/4275240`
  - **Reference:** (Fourati, 2020)

- **Dataset 7: Amazon Commerce Reviews**
  - **Description:** This dataset is designed for authorship identification using online writeprint, derived from customer reviews on Amazon Commerce Website. It includes linguistic style features for 50 of the most active users, each with 30 reviews.
  - **Characteristics:** Multivariate, Text, Domain-Theory. 1500 instances, 10000 features.
  - **Access:** UCI Machine Learning Repository.
  - **Dataset Link:** `https://archive.ics.uci.edu/dataset/215/amazon+commerce+reviews+set`
  - **Reference:** (Liu, 2011)

Table 3: Text Data Overview

Dataset & **Context** & **Target(s)** & **License** & **Annexure**
Tunizi & Arabizi social media & `sentiment` & CC BY-NC 4.0 & A-6
SA News Corpus & Setswana/Sepedi news & `topic` & CC BY-NC 4.0 & A-5
Amazon Reviews & E-commerce authorship & `author ID` & CC BY 4.0 & A-7

Compiled by the author from publicly available datasets.

Image Data The image-based datasets consist of grayscale and RGB image arrays with corresponding categorical targets. Each image undergoes a series of preprocessing steps to be transformed into a numeric tensor suitable for convolutional neural network (CNN) architectures, enhancing generalisation performance.

- **Dataset 8: MNIST (Image Classification)**

  - **Description:** A foundational dataset of 70,000 grayscale handwritten digits (0-9), size-normalized to 28x28 pixels. It is widely used for training and testing image classification models due to its simplicity and clear visual patterns.
  - **Characteristics:** Image classification data, with a 'train' split of 60,000 examples and a 'test' split of 10,000 examples. Each instance includes an `image` (28x28x1, uint8) and a `label` (int64, 10 classes).
  - **Access:** TensorFlow Datasets.
  - **Dataset Link:** `https://www.tensorflow.org/datasets/catalog/mnist`
  - **Reference:** (LeCun et al., 2010)

- **Dataset 9: Hands and Palm Images Dataset**
  - **Description:** A collection of 11,076 hand images (1600 x 1200 pixels) from 190 subjects (aged 18-75) primarily for gender and age detection. Images include both right and left hands from dorsal and palmar sides against a white background, accompanied by detailed metadata.
  - **Characteristics:** Image data suitable for digital image classification, particularly for biometric applications.
  - **Access:** Kaggle.
  - **Dataset Link:** `https://www.kaggle.com/datasets/shyambhu/hands-and-palm-images-data resource=download`
  - **Reference:** (Afifi, 2019)

Table 4: Image Data Overview

Dataset & **Use Case** & **Target(s)** & **License** & **Annexure**
MNIST (TFDS) & Digit classification & `label` (0–9) & CC BY 4.0 & A-8
11K Hands & Palms & `age`, `gender` & DbCL v1.0 & A-9

Compiled by the author from publicly available datasets.

Public Use Compliance

All nine datasets selected for this study are openly distributed under appropriate public-use licenses, including Creative Commons Attribution (CC BY 4.0), Creative Commons Attribution–NonCommercial (CC BY-NC 4.0), Database Contents License (DbCL v1.0), and API-based fair usage rights. None of the datasets contain personally identifiable information (PII), and all have been verified for non-commercial, research-grade access in alignment with South Africa's Protection of Personal Information Act (POPIA) and will be verified by the North-West University's ethics protocols.

For detailed information on source origin, licensing agreements, scope of permissible usage, de-identification statements, and verification links for each dataset, please refer to the

Public Use Compliance Records in the appendix (Annexures A-1 to A-9). These records will be submitted alongside the NWU ethics clearance bundle to demonstrate proactive adherence to open data licensing and institutional governance.

Data Preparation and Transformations

Data management and preparation are critical phases, ensuring that raw data from diverse modalities are transformed into a suitable format for machine learning models. This involves tailored preprocessing pipelines for structured, text, and image data, detailed below.

Structured Data Transformation Pipeline Structured datasets, characterized by numeric and categorical columns, undergo a standardized pipeline involving target alignment, encoding, and feature scaling.

1. **Target Alignment and Encoding:**
   - **Wine Quality:** The ordinal `quality` target (ranging from 3 to 9) is mapped into three distinct classes: `low`, `medium`, and `high`, enabling a multiclass classification approach.
   - **Credit Approval:** The binary target `A16` (`+`/`-`) is label-encoded into numerical values (e.g., 0 and 1).
   - **Yahoo Finance:** Historical closing prices are converted into categorical bins representing directional trends (e.g., increase, decrease, stagnant) relative to prior months, facilitating time-window-based classification. Tickers are integrated as covariates.
2. **Feature Encoding:** Categorical features across datasets are transformed using one-hot encoding to convert nominal variables into a numerical format suitable for model input.
3. **Numerical Scaling:** Numerical fields are normalized using Z-score standardisation, ensuring that features contribute equally to model training regardless of their original scale. The formula for Z-score standardisation is given by:

$$x' = \frac{x - \mu}{\sigma} \tag{7}$$

where (x=original value, $\mu = meanand\sigma = standarddeviation$).

Text-to-Numeric Transformation Pipeline Textual datasets require specific preprocessing to convert raw text into numerical representations (vectors) for machine learning models. This pipeline involves cleaning, tokenisation, and vectorisation.

1. **Text Pre-cleaning:** Raw text columns (e.g., `comment`, `review`, `text`) undergo a cleaning process including stopword removal, lowercasing, and punctuation stripping to reduce noise and standardize the text.

2. **Tokenisation:** Cleaned text is broken down into individual units (tokens), typically unigrams (single words) or bigrams (two-word sequences), depending on the granularity required for analysis.

3. **Vectorisation (TF-IDF Transformation):**

   - For datasets like **Tunizi**, **SA News Corpus**, and relevant aspect-based sentiment fields (`benefitsReview`, `sideEffectsReview`), a Term Frequency-Inverse Document Frequency (TF-IDF) transformation is applied. This technique captures the weighted lexical signals, indicating the importance of terms within a document relative to a corpus. The TF-IDF score for a term (t) in a document (d) from a corpus (D) is calculated as:

   $$\text{tfidf}(t, d, D) = \text{tf}(t, d) \times \text{idf}(t, D) \tag{8}$$

   where Term Frequency ((tf)) is:

   $$\text{tf}(t, d) = \frac{\text{count of } t \text{ in } d}{\text{number of words in } d} \tag{9}$$

   and Inverse Document Frequency ((idf)) is:

   $$\text{idf}(t, D) = \log\left(\frac{N}{|d' \in D : t \in d'|}\right) \tag{10}$$

   with (N) being the total number of documents in the corpus (D), and (—d' $\in D : t \in d'|)beingthenumberofdocumentsin(D)thatcontainterm(t)$.

   - For **Amazon Reviews**, a high-dimensional stylometric matrix is already pre-provided, thus no additional TF-IDF vectorisation is applied, as the data is already in a numerical feature format.

Image-to-Numeric Transformation Pipeline Image datasets, typically stored as raster files (e.g., `.png`, `.jpg`), necessitated a structured preprocessing pipeline to convert them into numeric tensors for Convolutional Neural Network (CNN) architectures. This process was implemented primarily via `tf.keras.utils.image_dataset_from_directory` and associated utilities for efficient data loading. Let $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ denote a source image, where H and W are spatial dimensions, and $C \in \{1, 3\}$ is the channel count.

The transformation sequence is as follows:

1. **Resising** Each image ($\mathbf{I}$) was resampled to a fixed spatial resolution ($\mathbf{I}_{\text{resized}} \in \mathbb{R}^{H' \times W' \times C}$) using bilinear interpolation.

   - **MNIST:** Retained at its native $28 \times 28 \times 1$ dimension.

- **11K Hands Dataset:** Downscaled from 1600×1200 to 224×224 pixels.

$$\mathbf{I}_{\text{resized}} = \text{Resize}(\mathbf{I}, H', W') \tag{11}$$

2. **Intensity Scaling** Pixel intensities were scaled to the range $[0, 1]$ by dividing by the maximum 8-bit value (255), denoted $\mathbf{I}_{\text{norm}}$. For transfer learning, channel-wise standardisation was applied, resulting in $\mathbf{I}_{\text{std}}$.

$$\mathbf{I}_{\text{norm}} = \frac{\mathbf{I}_{\text{resized}}}{255} \quad \text{and} \quad \mathbf{I}_{\text{std}} = \frac{\mathbf{I}_{\text{norm}} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad \boldsymbol{\mu}, \boldsymbol{\sigma} \in \mathbb{R}^C \tag{12}$$

3. **Augmentation (Stochastic)** To enhance generalisation and mitigate overfitting, domain-aware stochastic transformations ($\mathcal{T}$) were optionally applied to the training set.

$$\mathbf{I}_{\text{aug}} = \mathcal{T}(\mathbf{I}_{\text{std}}) \tag{13}$$

   The set of transformations included random rotation ($\mathcal{T}_{\text{rot}}(\theta)$), brightness adjustment ($\mathcal{T}_{\text{bright}}(\delta)$), occlusion (Cutout, $\mathcal{T}_{\text{occ}}(m, n)$), and horizontal flipping.

4. **Tensor Conversion** The final processed image was converted to the tensor format ($\mathbf{x} \in \mathbb{R}^{C \times H' \times W'}$) required for CNN input:

$$\mathbf{x} = \text{ToTensor}(\mathbf{I}_{\text{aug}}) \tag{14}$$

Analysis Tools

The analytical framework is built upon Python's robust ecosystem, employing a wide array of libraries for data handling, machine learning, deep learning, and advanced statistical analysis.

- **Data Manipulation and Preprocessing:**
  - **Pandas:** Fundamental for all data structuring, cleaning, transformation, and manipulation tasks, including DataFrame operations and data serialisation.
  - **NumPy:** Essential for numerical operations, array manipulation, and mathematical computations.
  - **re (Regular Expressions):** Used for advanced text cleaning and pattern matching in text-based data.
  - **zipfile** and **os:** For managing compressed archives and interacting with the file system, particularly for image data extraction.
  - **io:** For in-memory file handling during data import.
  - **scipy.io.arff:** For importing data from ARFF file format.
- **Machine Learning and Deep Learning Frameworks:**
  - **scikit-learn:** A comprehensive library for traditional machine learning models, including:

    - **Preprocessing:** `StandardScaler`, `LabelEncoder`.
    - **Model Selection:** `train`$_t est_s plit$, `KFold`, `RepeatedKFold`, `StratifiedKFold`, `LeavePOut`, `Gr`
    - **Clustering:** `KMeans`.
  - **TensorFlow/Keras:** For building and training deep learning models:
    - `Sequential` API for model definition.
    - `LSTM` (Long Short-Term Memory) layers for sequential data.
    - `Conv2D`, `MaxPooling2D`, `Flatten` layers for Convolutional Neural Networks (CNNs) on image or image-like data.
    - `Dense` layers for standard neural network layers.
  - `to`$_c ategorical for one - hot encoding target variables$.
  - **sentence-transformers:** For generating advanced text embeddings (e.g., `paraphrase-multilin`
  - **PyTorch** and **higher:** For implementing meta-learning algorithms like Reptile, leveraging automatic differentiation and inner-loop optimisation.

- **Model Evaluation and Reliability Analysis:**
  - **scikit-learn.metrics:** Provides a wide range of ML evaluation metrics like `accuracy, precision, recall, f1, roc-auc scores, confusion matrix, and matthews correlation coefficient`.
  - **scipy.stats:** provides a range of statistical evaluation tests, including `ks 2samp, (Kolmogorov-Smirnov test), chi`$^2 contingency(Chi - squared test), and F - oneway(ANOVA)$.
  - **scipy.special:** For mathematical functions like `rel-entr` (relative entropy for KL divergence).
  - **scipy.spatial.distance:** For distance metrics, specifically `jensenshannon` for JS divergence.
  - **joblib:** For efficient serialisation and deserialisation of Python objects, including DataFrames containing nested structures.
  - **shap:** For model explainability using SHAP (SHapley Additive exPlanations) values.
  - **lime:** For local model interpretability using LIME (Local Interpretable Model-agnostic Explanations).
  - **DoWhy** and **EconML:** Mentioned for their capabilities in causal inference, indicating a potential for future or advanced causal analysis within the research.

0.6.2 Motivation for Methodological Choices

The rigorous evaluation of the proposed statistical framework necessitates the strategic selection of specific advanced statistical and machine learning techniques. These choices are not arbitrary; they are driven by the existing literature gaps and the critical need to establish robust, statistically-sound procedures for trustworthy classification.

Data Modalities

The choice to test the framework across Structured, Text, and Image data modalities is a necessary condition for establishing the statistical rigour and broad utility of the final framework. These three modalities possess fundamentally different statistical properties, data representations (tabular features, sequential tokens, high-dimensional pixels), and inherent complexities Yoo et al. (2022). A robust statistical classification framework must demonstrate stability and efficacy across this heterogeneity. The core motivation for this selection is to move beyond modality-specific findings and generate a set of modality-optimal solutions (i.e., identifying the optimal combination of partitioning technique and model paradigm for *each* data type). This approach validates the framework's broad utility and directly addresses the need for modality-aware solutions in trustworthy AI.

Data Partitioning Techniques

The decision to benchmark a comprehensive set of techniques (up to 12) available through established libraries such as scikit-learn is a foundational methodological choice. Data partitioning is not a one-size-fits-all problem; the inherent statistical structure of different data modalities interacts uniquely with simple random, stratified, group-based, and time-series splits, leading to varying statistical outcomes Tan et al. (2021). This systematic approach allows the study to move beyond assumptions and empirically determine which partitioning strategy yields the highest statistical stability and generalisability for *each* specific modality. This rigorous, modality-sensitive benchmarking is essential to creating a robust statistical framework.

Traditional ML and Meta-Learning Model Paradigms

The study's comparative analysis is structured around two distinct model paradigms: Traditional Machine Learning (ML) and Meta-Learning. Traditional ML models serve as essential, well-understood baselines, providing a measure of established performance and robustness. In contrast, Meta-Learning Models (e.g., Reptile, Stacking Meta-Learner) are explicitly chosen for their advanced capacity to 'learn how to learn' from diverse tasks and adapt rapidly to new, shifted data distributions Hospedales et al. (2020). The comparison directly tests whether the added statistical complexity and inherent adaptability of Meta-Learning is empirically and statistically justified over robust Traditional ML when applied across the three data modalities.

Reliability Analysis Metrics

The selection and integration of advanced reliability metrics represent the final, statistically rigorous component of the framework. Relying solely on predictive accuracy is

statistically insufficient for trustworthy AI and responsible development Suresh and Ghassemi (2021). Our chosen metrics address the three pillars of model trustworthiness:

- **Statistical Stability (Drift):** Metrics like the Kolmogorov-Smirnov Test and Population Stability Index (PSI) are vital for quantifying the degree of data or concept drift, ensuring the model's statistical premises remain valid over time.
- **Ethical Equity (Fairness):** Indicators such as Statistical Parity and Equalized Odds quantify bias across sensitive subgroups, ensuring the model's predictions are ethically and equitably distributed.
- **Transparency (Explainability):** Tools like SHAP and Causality Checks provide transparency into the decision-making process, converting the model from a "black box" into a statistically interpretable tool Arrieta et al. (2020).

These metrics are not ancillary; they serve as the primary statistical criteria for the final model recommendation, ensuring the transition from framework development to model selection is rigorous, stable, and trustworthy.

0.6.3 Data Partitioning Techniques

Standard Holdout - Techniques Standard holdout methods are foundational strategies for partitioning datasets into mutually exclusive training and testing subsets. These methods assume that samples are independently and identically distributed (i.i.d.) and are often used as baselines for evaluating model performance (Hastie et al., 2009). Two primary variants are discussed in this section: **Random Split** and **Stratified Split**.

Random Split. The Random Split approach selects a fraction of the full dataset D=(x i ,y i) i=1 n to be held out as the test set. This sampling is performed uniformly at random and does not consider the label distribution. The resulting training and testing datasets can be defined as:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{RandomSplit}(\mathcal{D}, \theta) \tag{15}$$

Stratified Split. To address issues of label imbalance in classification tasks, Stratified Split ensures that the class proportions are preserved across the training and test sets (Kohavi, 1995). For each class cC, the ratio of samples is maintained:

$$\frac{|\mathcal{D}\text{train}^c|}{|\mathcal{D}\text{train}|} \approx \frac{|\mathcal{D}^c|}{|\mathcal{D}|} \quad \text{and} \quad \frac{|\mathcal{D}\text{test}^c|}{|\mathcal{D}\text{test}|} \approx \frac{|\mathcal{D}^c|}{|\mathcal{D}|} \tag{16}$$

Statistical Assumptions. Both splitting methods assume:

- Each observation is i.i.d. and exchangeable.
- The dataset is sufficiently large to approximate the distribution in both subsets.
- In the case of stratification, each class label has a sufficient number of instances to permit proportional splitting.

Pseudo-code Presentation. Illustration of train test splitter for Standard holdout:

```
FUNCTION StandardHoldoutSplit(X, y, θ, stratify_flag):

    IF stratify_flag IS TRUE:
        GROUP data (X, y) BY class labels
        FOR each class c IN class set C:
            RANDOMLY SELECT (θ × number of instances in class c) AS test set
            ASSIGN remaining instances in class c TO train set
    ELSE:
        SHUFFLE all instances uniformly at random
        SELECT (θ × total number of instances) AS test set
        ASSIGN remaining instances TO train set

    RETURN X_train, X_test, y_train, y_test
```

Figure 2: Random and Stratified Splitting Pseudo-code (Author's conceptualisation)

Cross-Validation - Techniques Cross-validation methods improve model reliability by evaluating performance across multiple training and testing partitions. These techniques ensure robustness, prevent overfitting, and provide a more accurate estimate of generalisation error (Hastie et al., 2009).

K-Fold Cross-Validation. K-Fold Cross-Validation assesses model performance by dividing the dataset into k equally sized folds. Each fold is used as a test set once, while the remaining k1 folds serve as training data. The final performance estimate is obtained by averaging across all folds.

Mathematical Definition. Let D be the dataset, split into k partitions:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{KFoldSplit}(\mathcal{D}, k) \tag{17}$$

Statistical Assumptions.

- Instances are independently and identically distributed (i.i.d.).

- The dataset is sufficiently large to allow effective partitioning.

Pseudo-code Presentation. Illustration of the K-Fold splitting technique.

```
FUNCTION KFoldSplit(X, y, k):

    SHUFFLE dataset randomly
    SPLIT dataset into k equally sized folds

    FOR each fold i IN {1, ..., k}:
        ASSIGN fold i AS test set
        ASSIGN remaining k-1 folds AS training set

        RETURN X_train[i], X_test[i], y_train[i], y_test[i]
```

Figure 3: K-Fold Cross-Validation Pseudo-code (Author's conceptualisation)

Repeated K-Fold Cross-Validation. Repeated K-Fold Cross-Validation extends K-Fold by performing multiple iterations with different random partitions, reducing variance in the final performance estimate.

Mathematical Definition. Let D be the dataset, split into k folds across r repetitions:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{RepeatedKFoldSplit}(\mathcal{D}, k, r) \tag{18}$$

Statistical Assumptions.

- Instances are exchangeable across repetitions.
- Sufficient dataset size to support repeated resampling.

Pseudo-code Presentation. Illustration of the Repeated K-Fold splitting technique.

```
FUNCTION RepeatedKFoldSplit(X, y, k, num_repeats):

    FOR each repetition r IN {1, ..., num_repeats}:
        SHUFFLE dataset randomly
        SPLIT dataset into k equally sized folds

        FOR each fold i IN {1, ..., k}:
            ASSIGN fold i AS test set
            ASSIGN remaining k-1 folds AS training set

            RETURN X_train[r][i], X_test[r][i], y_train[r][i], y_test[r][i]
```

Figure 4: Repeated K-Fold Cross-Validation Pseudo-code (Author's conceptualisation)

Stratified K-Fold Cross-Validation. Stratified K-Fold ensures that class proportions are preserved across training and testing sets, which is crucial when working with imbalanced data (Kohavi, 1995).

Mathematical Definition. Given dataset D with class labels C, stratified splitting maintains:

$$\frac{|\mathcal{D}\text{train}^c|}{|\mathcal{D}\text{train}|} \approx \frac{|\mathcal{D}^c|}{|\mathcal{D}|}, \quad \frac{|\mathcal{D}\text{test}^c|}{|\mathcal{D}\text{test}|} \approx \frac{|\mathcal{D}^c|}{|\mathcal{D}|} \tag{19}$$

Statistical Assumptions.

- Each class label appears sufficiently in both training and testing sets.
- Class proportions remain stable across splits.

Pseudo-code Presentation. Illustration of the Stratified K-Fold splitting technique.

```
FUNCTION StratifiedKFoldSplit(X, y, k):

    GROUP data (X, y) BY class labels
    SPLIT each class subset into k equally sized folds

    FOR each fold i IN {1, ..., k}:
        ASSIGN fold i AS test set while maintaining class proportions
        ASSIGN remaining k-1 folds AS training set

        RETURN X_train[i], X_test[i], y_train[i], y_test[i]
```

Figure 5: Stratified K-Fold Cross-Validation Pseudo-code (Author's conceptualisation)

Repeated Stratified K-Fold Cross-Validation. Repeated Stratified K-Fold applies multiple iterations of stratified partitioning to improve model performance stability.

Mathematical Definition. Given dataset D with class labels C, repeated stratified partitioning maintains:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{RepeatedStratifiedKFoldSplit}(\mathcal{D}, k, r) \tag{20}$$

Pseudo-code Presentation. Illustration of the Repeated Stratified K-Fold splitting technique.

```
FUNCTION RepeatedStratifiedKFoldSplit(X, y, k, num_repeats):

    FOR each repetition r IN {1, ..., num_repeats}:
        GROUP data (X, y) BY class labels
        SPLIT each class subset into k folds while
        preserving class distribution

        FOR each fold i IN {1, ..., k}:
            ASSIGN fold i AS test set while maintaining
            class proportions
            ASSIGN remaining k-1 folds AS training set

            RETURN X_train[r][i], X_test[r][i],
                   y_train[r][i], y_test[r][i]
```

Figure 6: Repeated Stratified K-Fold Cross-Validation Pseudo-code (Author's conceptualisation)

Leave-P-Out. Leave-P-Out selects P samples per iteration as the test set, ensuring the remaining dataset is used for training. This method is exhaustive for small datasets but computationally demanding.

Mathematical Definition. Given dataset D with n instances, Leave-P-Out defines:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{LeavePOutSplit}(\mathcal{D}, P) \tag{21}$$

Pseudo-code Presentation. Illustration of the Leave-P-Out splitting technique.

```
FUNCTION LeavePOutSplit(X, y, P):

    DETERMINE total instances n = size of dataset
    IDENTIFY P = number of instances to leave out per iteration

    GENERATE all possible subsets of P instances as test sets
    FOR each subset S:
        ASSIGN S AS test set
        ASSIGN remaining (n-P) instances AS training set

        RETURN X_train[S], X_test[S], y_train[S], y_test[S]
```

Figure 7: Leave-P-Out Cross-Validation Pseudo-code (Author's conceptualisation)

Shuffle-Split Variants - Techniques Shuffle-Split methods provide an alternative to traditional cross-validation techniques by randomly partitioning the dataset multiple times, ensuring flexibility in train-test proportions. Stratified ShuffleSplit extends this approach by maintaining class distributions across partitions.

Stratified ShuffleSplit. Stratified ShuffleSplit ensures that each randomized split preserves the proportion of samples per class label, addressing imbalances in classification tasks (Pedregosa et al., 2011).

Mathematical Definition. Given dataset D with class labels C, stratified partitioning maintains:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{StratifiedShuffleSplit}(\mathcal{D}, s, \theta) \tag{22}$$

Pseudo-code Presentation. Illustration of the Stratified ShuffleSplit technique.

```
FUNCTION StratifiedShuffleSplit(X, y, num_splits, test_fraction):

    FOR each split s IN {1, ..., num_splits}:
        GROUP data (X, y) BY class labels
        SHUFFLE instances within each class randomly
        SELECT (test_fraction × total instances per class) AS test set
        ASSIGN remaining instances AS training set

        RETURN X_train[s], X_test[s], y_train[s], y_test[s]
```

Figure 8: Stratified ShuffleSplit Pseudo-code (Author's conceptualisation)

Group-Aware - Techniques Group-aware methods ensure that entire groups remain together within either the training or test set, preventing data leakage and ensuring realistic

evaluation scenarios. These methods are particularly useful when working with clustered data or subject-based observations (Pedregosa et al., 2011).

Group K-Fold. Group K-Fold partitioning ensures that all instances of a given group remain within a single fold during cross-validation. This prevents data leakage and maintains independence across training and test subsets.

Mathematical Definition. Given dataset D with group labels G, the split ensures:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{GroupKFoldSplit}(\mathcal{D}, \mathcal{G}, k) \tag{23}$$

Pseudo-code Presentation. Illustration of the Group K-Fold splitting technique.

```
FUNCTION GroupKFoldSplit(X, y, groups, k):

    IDENTIFY unique groups in dataset
    SPLIT dataset into k folds while maintaining group integrity

    FOR each fold i IN {1, ..., k}:
        ASSIGN fold i AS test set ensuring no group overlap
        ASSIGN remaining k-1 folds AS training set

        RETURN X_train[i], X_test[i], y_train[i], y_test[i]
```

Figure 9: Group K-Fold Cross-Validation Pseudo-code (Author's conceptualisation)

Group ShuffleSplit. Group ShuffleSplit extends traditional ShuffleSplit by ensuring that all instances within a group remain together while still allowing randomized partitions across multiple splits.

Mathematical Definition. Given dataset D with group labels G, the split process follows:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{GroupShuffleSplit}(\mathcal{D}, \mathcal{G}, s, \theta) \tag{24}$$

Pseudo-code Presentation. Illustration of the Group ShuffleSplit technique.

```
FUNCTION GroupKFoldSplit(X, y, groups, k):

    IDENTIFY unique groups in dataset
    SPLIT dataset into k folds while maintaining group integrity

    FOR each fold i IN {1, ..., k}:
        ASSIGN fold i AS test set ensuring no group overlap
        ASSIGN remaining k-1 folds AS training set

        RETURN X_train[i], X_test[i], y_train[i], y_test[i]
```

Figure 10: Group ShuffleSplit Pseudo-code (Author's conceptualisation)

Stratified Group K-Fold. Stratified Group K-Fold ensures that class proportions are maintained while preserving group integrity across cross-validation folds.

Mathematical Definition. Given dataset D with group labels G and class labels C, stratified partitioning maintains:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} \sim \text{StratifiedGroupKFoldSplit}(\mathcal{D}, \mathcal{G}, \mathcal{C}, k) \tag{25}$$

Pseudo-code Presentation. Illustration of the Stratified Group K-Fold splitting technique.

```
FUNCTION StratifiedGroupKFoldSplit(X, y, groups, k):

    GROUP data (X, y) BY class labels while
    maintaining group integrity

    SPLIT dataset into k folds while preserving
    both class balance & group structure

    FOR each fold i IN {1, ..., k}:
        ASSIGN fold i AS test set ensuring
        class proportions & no group overlap

        ASSIGN remaining k-1 folds AS training set

        RETURN X_train[i], X_test[i],
               y_train[i], y_test[i]
```

Figure 11: Stratified Group K-Fold Cross-Validation Pseudo-code (Author's conceptualisation)

Reinforcement Learning-Based Partitioning RLAPT-Q Learning applies reinforcement learning principles to data partitioning, dynamically adjusting the split strategy based

on learned rewards. This approach optimizes the partitioning process, improving model generalisation.

RLAPT-Q Learning Framework. The RLAPT-Q framework integrates a reinforcement learning agent that selects partitioning actions based on state transitions and reward feedback. The Q-learning algorithm refines these decisions iteratively (Sutton and Barto, 1998).

Step 1: Initialisation. The Q-learning process begins by initialising the Q-table, which tracks state-action values. Each state corresponds to a dataset partitioning scenario, while each action represents a split decision.

```
FUNCTION RLAPT_Q_Initialize(state_space, action_space):

    DEFINE Q-table (state_space × action_space),
    initializing Q-values to small random values

    SET learning rate α, discount factor γ &
    exploration strategy (e.g., ε-greedy)

    RETURN initialized Q-table
```

Figure 12: RLAPT-Q Learning Step 1: Initialisation (Author's conceptualisation)

The Q-table is initialized as:

$$Q(s, a) = 0 \quad \forall s \in S, a \in A \tag{26}$$

where S represents all possible partition states and A denotes available splitting actions.

Step 2: Policy Execution. The agent selects actions using an exploration strategy, balancing between random partitioning and exploiting prior knowledge.

```
FUNCTION RLAPT_Q_Policy(state, Q_table, ε):

    WITH probability ε, SELECT random action
    from action_space

    OTHERWISE, SELECT action with highest
    Q-value for the given state

    RETURN selected action
```

Figure 13: RLAPT-Q Learning Step 2: Policy Execution (Author's conceptualisation)

The action selection follows an -greedy approach (Sutton and Barto, 1998):

$$a_t = \begin{cases} \text{random action}, & \text{with probability } \epsilon \\ \text{otherwise} & \arg\max_a Q(s_t, a), \end{cases} \tag{27}$$

where determines the exploration rate.

Step 3: Q-Value Update. After executing an action, the agent receives a reward based on partitioning efficiency and updates the Q-values accordingly.

```
FUNCTION RLAPT_Q_Update(Q_table, state, action,
                        reward, next_state):

    IDENTIFY max Q-value for next_state

    APPLY Bellman Equation:
    Q(state, action) ← Q(state, action) + α ×
    [reward + γ × max Q(next_state) - Q(state, action)]

    RETURN updated Q_table
```

Figure 14: RLAPT-Q Learning Step 3: Q-Value Update (Author's conceptualisation)

The Q-value update follows the standard Bellman equation (Sutton and Barto, 1998):

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max Q(s', a') - Q(s, a)] \tag{28}$$

where:

- is the learning rate.

- is the discount factor.
- r is the reward obtained.
- s represents the next state.

Step 4: Full Learning Loop. The reinforcement learning process iterates through multiple episodes, refining partitioning efficiency.

```
FUNCTION RLAPT_Q_Learning(state_space, action_space,
                          episodes):

    INITIALIZE Q-table

    FOR each episode e IN {1, ..., episodes}:
        RESET environment & SET initial state

        WHILE episode is active:
            SELECT action using RLAPT_Q_Policy(
                   state, Q_table, ε)

            EXECUTE action, OBSERVE reward,
            next_state

            UPDATE Q-table using RLAPT_Q_Update(
                   Q_table, state, action, reward,
                   next_state)

            MOVE to next_state

    RETURN optimized Q-table
```

Figure 15: RLAPT-Q Learning Step 4: Full Learning Loop (Author's conceptualisation)

The iterative learning follows:

$$\text{for each episode:} \quad s_0 \rightarrow a_t \rightarrow r_t \rightarrow s_{t+1} \tag{29}$$

until convergence criteria are met, ensuring optimal partitioning decisions.

Adaptive Train-Test Partitioning (ATTP) - Proposed Method Adaptive Train-Test Partitioning (ATTP) introduces a methodological framework that systematically refines dataset partitioning by leveraging dimensionality reduction, clustering, stratified sampling, and iterative optimisation. This approach enhances model generalisation and minimizes biases in train-test splits.

Dimensionality Reduction: Linear Discriminant Analysis (LDA). LDA transforms high-dimensional feature spaces into lower-dimensional representations while maximising class separability. This step ensures that partitioning retains discriminative features (Fisher, 1936; Hastie et al., 2009).

Mathematical Definition. Let X be the feature matrix and y the class labels. LDA optimizes:

$$W = \arg\max_{W} \frac{|W^T S_B W|}{|W^T S_W W|} \tag{30}$$

where S B

represents the between-class scatter matrix and S W is the within-class scatter matrix.

Pseudo-code Presentation. Illustration of the Linear Discriminant Analysis (LDA) step.

```
FUNCTION ApplyLDA(X, y, num_components):

    COMPUTE mean vectors per class label

    CALCULATE within-class scatter matrix Sw,
    between-class scatter matrix Sb

    SOLVE eigenvalue problem for Sw⁻¹ Sb &
    select leading eigenvectors

    RETURN transformed dataset
```

Figure 16: ATTP Step 1: Dimensionality Reduction using LDA (Author's conceptualisation)

Unsupervised Clustering: k-Means. Applying k-Means clustering on the reduced data helps identify natural sub-groups within the dataset, ensuring meaningful segmentation (MacQueen, 1967; Hastie et al., 2009).

Mathematical Definition. Given feature matrix X after LDA transformation, k-Means optimizes:

$$\arg\min_{C} \sum_{i=1}^{k} \sum_{x \in C_i} |x - \mu_i|^2 \tag{31}$$

where C i

represents cluster assignments and i

denotes centroid positions.

Pseudo-code Presentation. Illustration of the k-Means clustering step.

```
FUNCTION ApplyKMeans(X_reduced, num_clusters):

    INITIALIZE num_clusters centroids randomly

    REPEAT until convergence:
        ASSIGN instances to nearest centroid

        UPDATE centroids using mean position
        of assigned instances

    RETURN cluster labels & final centroids
```

Figure 17: ATTP Step 2: k-Means clustering on reduced data (Author's conceptualisation)

Stratified Sampling. Once clusters are identified, stratified sampling ensures proportional representation within both training and test partitions.

Mathematical Definition. For each class c in clustered data C:

$$\frac{|\mathcal{D}\text{train}^c|}{|\mathcal{D}\text{train}|} \approx \frac{|\mathcal{D}^c|}{|\mathcal{D}|}, \quad \frac{|\mathcal{D}\text{test}^c|}{|\mathcal{D}\text{test}|} \approx \frac{|\mathcal{D}^c|}{|\mathcal{D}|} \tag{32}$$

Pseudo-code Presentation. Illustration of the stratified sampling step.

```
FUNCTION StratifiedSampling(X, y, sample_fraction):

    GROUP data (X, y) BY class labels

    FOR each class c:
        SELECT (sample_fraction × total instances
        in class c) AS sample set

    RETURN stratified sampled dataset
```

Figure 18: ATTP Step 3: Stratified Sampling (Author's conceptualisation)

Accuracy Evaluation. Partitioning efficiency is evaluated using accuracy metrics, ensuring test subsets reflect expected distributions (Hastie et al., 2009).

Mathematical Definition. Accuracy is defined as:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}} \times 100 \tag{33}$$

Pseudo-code Presentation. Illustration of the accuracy evaluation step.

```
FUNCTION ComputeAccuracy(y_true, y_pred):

    COUNT correct predictions
    CALCULATE accuracy as (correct_predictions / total_samples) × 100%

    RETURN accuracy percentage
```

Figure 19: ATTP Step 4: Accuracy Evaluation (Author's conceptualisation)

Iterative Optimisation. Finally, iterative optimisation refines partitioning by adjusting train-test ratios based on performance feedback.

Mathematical Definition. Optimisation iterates as:

$$\mathcal{D}\text{train}, \mathcal{D}\text{test} = \text{Optimize}(\mathcal{D}, \theta, \mathcal{M}) \tag{34}$$

where M represents learned partitioning metrics.

Pseudo-code Presentation. Illustration of the iterative optimisation step.

```
FUNCTION IterativeOptimization(model, X_train,
                               y_train, max_iterations):

    FOR each iteration i IN {1, ..., max_iterations}:
        UPDATE model using optimization algorithm

        VALIDATE improvement

        STOP if convergence criteria met

    RETURN optimized model parameters
```

Figure 20: ATTP Step 5: Iterative Optimisation (Author's conceptualisation)

ATTP Process Overview. To summarize, the ATTP methodology combines multiple statistical approaches—LDA for dimensionality reduction, k-Means for clustering, stratified sampling for balanced splits, accuracy evaluation for partition validation, and iterative optimisation for continuous refinement.

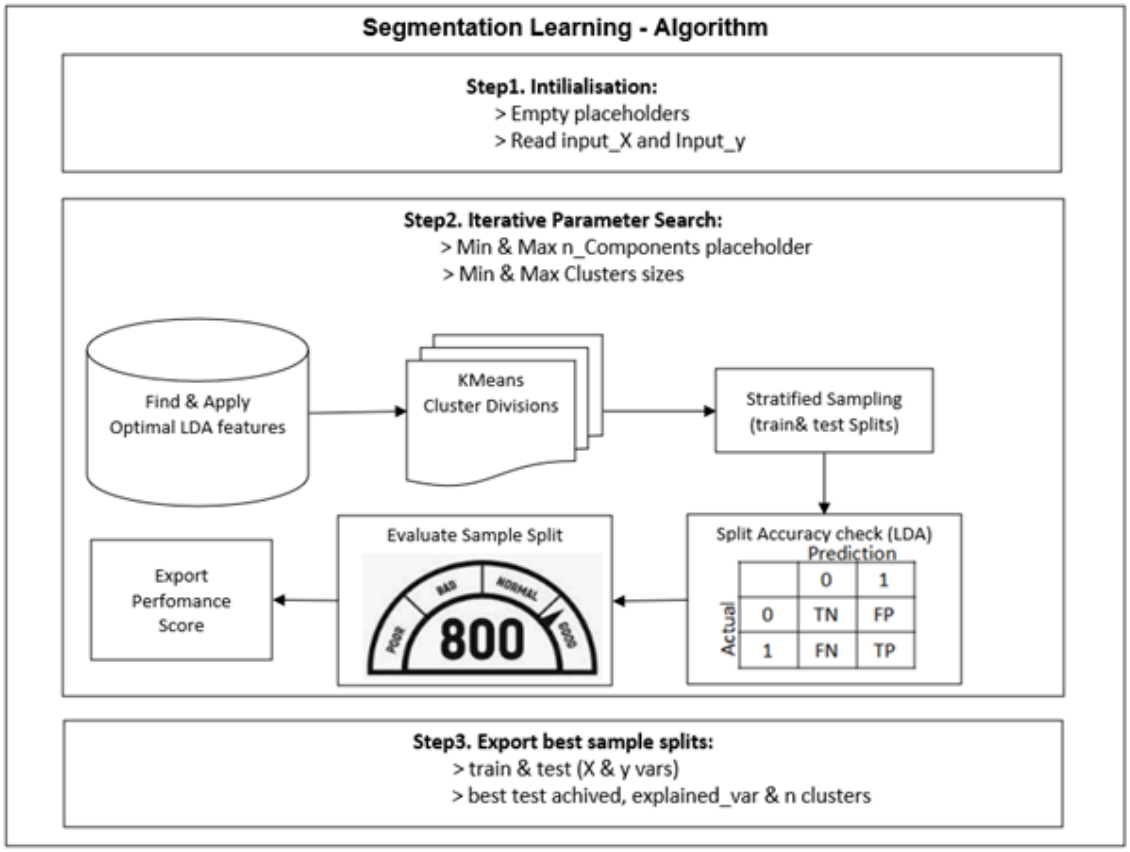


Figure 21: ATTP Process-Overview of all methodological steps (Author's conceptualisation)

0.6.4 Traditional ML and Meta-Learning Models

Machine learning models are fundamental to reliability assessments across various data types. This section details traditional, deep learning, and meta-learning models, highlighting their suitability and perceived strengths based on data modalities.

Traditional ML Models Traditional machine learning models are widely used for their interpretability, robustness, and computational efficiency across diverse datasets.

Random Forest (RF) **Justification:** A robust ensemble method known for its ability to handle complex interactions, analyze feature importance, and generalize well across different data distributions (Breiman, 2001). Its perceived strength lies in its effectiveness with **structured/tabular data** due to its ability to capture non-linear relationships and interactions between features, making it a strong performer in various predictive tasks,

including finance and healthcare applications (IBM, ndb).

Pros:

- High accuracy and less prone to overfitting compared to single decision trees (Breiman, 2001).
- Handles high-dimensional data, effective with missing values and imbalanced datasets (Breiman, 2001; IBM, ndb).

Cons:

- Can be computationally intensive for very large datasets and may have slower inference times (IBM, ndb).
- Less interpretable than single decision trees due to its ensemble nature (IBM, ndb).

Mathematical Definition. RF constructs multiple decision trees (T_i)$onbootstrappeddatasetsandaggrego$ $\frac{1}{M}\sum_{i=1}^{M} T_i(X)(35)$ where:

- (M) is the number of trees in the forest.
- (X) is the input feature set.
- ($T_i(X)$)$isthepredictionfromthe(i^{th})tree.$

Reference: Breiman (2001).

Logistic Regression (LR) **Justification:** A fundamental statistical model providing interpretable probability estimations, often serving as a reliable baseline for classification tasks (Cox, 1958). Its perceived strength is in its utility for **structured/tabular data**, particularly for binary classification, due to its simplicity, interpretability of feature importance, and ability to output probabilities, suitable for applications like fraud detection, disease prediction, and churn analysis (IBM, nda; Coursera, nd).

Pros:

- Computationally inexpensive and provides easily interpretable coefficients (IBM, nda).
- Effective for linearly separable datasets and can handle both numeric and categorical input variables (Coursera, nd).

Cons:

- Struggles with non-linear relationships and can be sensitive to multicollinearity.
- May perform poorly with high feature redundancy.

Mathematical Definition. LR estimates the probability that (X) belongs to class (y = 1) using the sigmoid function:

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta^T X)}} \tag{36}$$

where:

- $(\ \beta_0) is the bias term. (\beta) represents feature weights.$
- (X) denotes the input vector.

Reference: Cox (1958).

Naïve Bayes (NB) Justification: A probabilistic classifier well-suited for text and structured data due to its simplicity, speed, and effectiveness under the assumption of feature independence (Lewis, 1998). Its perceived strength lies particularly in **text data** and **sparse data** like bag-of-words representations, making it highly effective for text classification tasks such as spam detection, sentiment analysis, and topic modelling, even if its strong independence assumption is often violated in practice (Analytics, nda; RapidMiner, nd).

Pros:

- Excellent speed and efficiency, robustness on small datasets (RapidMiner, nd).
- Effective for high-dimensional data, especially with sparse feature spaces.

Cons:

- The strong assumption of feature independence often doesn't hold true in real-world data, potentially limiting performance in complex datasets (Analytics, nda).
- Can suffer from the "zero-frequency problem" where a zero probability is assigned if a feature value is not present in the training data, which can be mitigated with Laplace smoothing (RapidMiner, nd).

Mathematical Definition. NB estimates the probability of a given data instance belonging to class (y) using Bayes' theorem:

$$P(y|X) = \frac{P(X|y)P(y)}{P(X)} \tag{37}$$

where:

- (P(X $\|y$)) $is the likelihood of features given class (y). (P(y)) is the prior probability of the class.$

- (P(X)) is the probability of the observed feature set.

Reference: Lewis (1998).

Support Vector Machine (SVM) **Justification:** A powerful discriminative classifier that constructs hyperplanes in high-dimensional space for clear separation between classes, suitable for both classification and regression (Cortes and Vapnik, 1995). Its perceived strength lies in its effectiveness with **high-dimensional data**, including **image and text features**, where it can find clear decision boundaries even in complex datasets by transforming data into higher dimensions using various kernel functions (Keylabs, nd; Singh, 2025).

Pros:

- Effective in high-dimensional spaces, good generalisation even with limited training data (Singh, 2025).
- Robust against overfitting due to the maximisation of the margin (Singh, 2025).
- Versatile with different kernel functions (e.g., linear, polynomial, RBF) to handle non-linear data (Singh, 2025).

Cons:

- Can be computationally intensive and slow for very large datasets due to the quadratic programming problem involved.
- Performance is highly dependent on the choice of kernel function and regularisation parameters, requiring careful tuning (Singh, 2025).

Mathematical Definition. SVM classifies an input feature (X) by maximising the margin between classes, solving the optimisation problem:

$$\min_{w} \sum_{i} \max(0, 1 - y_i(w^T X_i + b)) \tag{38}$$

where:

- (X_i) $is an input feature vector. (y_i) is the class label.$

- (w) and (b) define the decision boundary.

Reference: Cortes and Vapnik (1995).

Deep Learning Models Deep learning models excel at automatically learning hierarchical features from complex data, particularly in sequence and spatial pattern recognition tasks.

Long Short-Term Memory (LSTM) **Justification:** A type of Recurrent Neural Network (RNN) specifically designed to capture sequential dependencies and long-range patterns in data, making it ideal for time series and text processing (Hochreiter and Schmidhuber, 1997). Its perceived strength is its efficacy with **sequential data**, including **time series** and **text**, due to its memory cells and gating mechanisms that control information flow over extended periods, crucial for tasks like language modelling, machine translation, and time series forecasting (GeeksforGeeks, ndb; Firdaus, 2023).

Pros:

- Effectively handles vanishing/exploding gradient problems inherent in traditional RNNs (GeeksforGeeks, ndb).
- Capable of learning long-term dependencies in sequential data.

Cons:

- Computationally expensive to train, especially for very long sequences.
- Its internal mechanisms (gates) can make interpretation challenging.

Mathematical Definition. LSTM units compute hidden states (h_t)$ateachtimestepusingcomplexgatingn$
$\sigma(W_h h_{t-1} + W_x X_t + b_h)(39)$ where:

- (W_h, W_x)$areweightmatrices.$(X_t)$representsthecurrentinput.$

- (h_{t-1})$isthepreviousshiddenstate.$($b_h$)$isthebiasterm.$
- (σ)$denotesanactivationfunction(note : thissimplifiedequationdoesn't capturethefullgatemechanismbutr$

Reference: Hochreiter and Schmidhuber (1997).

Convolutional Neural Networks (CNN) **Justification:** Industry-standard architectures for processing data with a known grid-like topology, such as images, by automatically learning spatial hierarchies of features (LeCun et al., 1998). Its perceived strength lies in its exceptional performance with **image data** and other **grid-like data**, making it the de facto standard for visual data processing (images, videos) due to its ability to learn hierarchical spatial features through convolutional layers and pooling, requiring minimal pre-processing (Wikipedia, nd; Islam and Islam, 2023).

Pros:

- Highly effective in image and spatial data recognition, capable of automatic feature learning (GeeksforGeeks, nda).

- Achieves state-of-the-art performance on large-scale datasets for tasks like image classification, object detection, and image segmentation (Wikipedia, nd; Islam and Islam, 2023).

Cons:

- Requires substantial computational resources and large amounts of data for effective training (GeeksforGeeks, nda).
- The learned features can be difficult to interpret directly.

Mathematical Definition. CNNs apply convolutional operations (W) over input data (X) to capture spatial dependencies, typically followed by an activation function:

$$X^{(l+1)} = \sigma(W^{(l)} * X^{(l)} + b^{(l)}) \tag{40}$$

where:

- (W$^{(l)}$)$represents learnable filter weights at layer (l).(b^{(l)}) is the bias term.$
- ($*$)$denotes the convolution operator.(\sigma) is the activation function.$

Reference: LeCun et al. (1998).

Meta-Learning Models Meta-learning, or "learning to learn," focuses on training models to adapt quickly to new tasks or distributions with minimal data.

Reptile Justification: A lightweight, gradient-based meta-learning algorithm designed for rapid adaptation, optimising initial model parameters so they can quickly generalize to new, unseen tasks with a few gradient steps (Nichol et al., 2018a). Its perceived strength lies in its efficiency for **few-shot learning** across **diverse task adaptations** and potentially various data types, enabling models to learn from limited examples by finding good initial parameters (OpenAI, 2018; AIMultiple, nd).

Pros:

- Efficient adaptation without extensive retraining, computationally less demanding than some other meta-learning algorithms (e.g., MAML) (OpenAI, 2018).
- Effective for few-shot learning scenarios.

Cons:

- May offer limited improvements compared to heavily fine-tuned traditional models on specific tasks.
- Its performance can be sensitive to the choice of inner and outer learning rates.

Mathematical Definition. Reptile updates the model parameters (θ) by iteratively optimising across multiple task distributions. This method aims to move the parameters toward the average of the task-specific optimal parameters (θ_i^*):

$$\theta \leftarrow \theta - \alpha \sum_{i=1}^{N} (\theta - \theta_i^{)} \tag{41}$$

where (α) is the meta-learning rate, (N) is the number of sampled tasks, and (θ_i^*) is the optimized parameter for task (i). The update is implemented using a gradient-based approximation, calculated as:($\theta \leftarrow \theta - \alpha \sum_{i=1}^{N} \nabla_\theta \mathcal{L}_i$)).

Reference: Nichol et al. (2018a).

PerfMetaSelect (Performance-Aware Meta-Model Selector) **Justification:** This approach embodies the concept of automatically selecting the best-performing base model from a set of candidates for a given task or dataset, often by training a meta-classifier on the performance metrics of base models (Feurer et al., 2019). Its perceived strength, as a component of **Automated Machine Learning (AutoML)**, is its ability to adaptively select optimal models or pipelines for **diverse data types** (structured, image, text, time-series) by iteratively trying various algorithms and parameters, making ML more accessible and efficient for complex real-world problems (Learn, nd; Nawaz and Shah, 2025).

Pros:

- Automates model selection, potentially leading to higher overall performance by dynamically choosing the most suitable model (Learn, nd; Nawaz and Shah, 2025).
- Can integrate various performance metrics beyond just accuracy.

Cons:

- Requires training multiple base models and a meta-model, increasing computational overhead.
- The effectiveness relies on the meta-model's ability to accurately predict which base model will perform best.
- Can be complex to interpret the final decision process (Nawaz and Shah, 2025).

Mathematical Definition. While there is no single universal mathematical definition for "PerfMetaSelect" as it describes an architectural concept, the framework typically involves a meta-learner (g) that predicts the most suitable base model (f_k) from a pool of candidates ($f_1, \ldots, f_K$). The selection is based on observed performance metrics ($\mathbf{P}$)

(e.g., accuracy, precision, recall) of these base models on a given dataset, with the primary objective being the maximisation of overall performance:

$$\text{Select } f_k \text{ s.t. } k = \arg\max_j \text{Metric}(f_j(\mathbf{X}_{\text{test}}), \mathbf{Y}_{\text{test}}) \tag{42}$$

The meta-learner then attempts to learn this optimal selection strategy.

Reference: Feurer et al. (2019).

Stacking Meta-Learner (Stacked Generalisation) Justification: An ensemble meta-learning technique that combines the predictions of multiple diverse base models using a "meta-learner" to make the final prediction, often leading to improved predictive performance (Wolpert, 1992). Its perceived strength is for **heterogeneous data** and **complex problems** where different base models capture different aspects of the data, making it a powerful technique across various domains and often leading to winning performance in competitions (Analytics, ndb; Veiga et al., 2024).

Pros:

- Can significantly improve predictive accuracy over individual base models by combining their strengths (Analytics, ndb).
- Robust across various types of datasets and problems by leveraging different learning algorithms (Veiga et al., 2024).

Cons:

- Can be computationally intensive due to training multiple base models and a meta-model (Analytics, ndb).
- The final combined model can be less interpretable than individual models (Veiga et al., 2024).

Mathematical Definition. Given a set of base models $(f_1, \ldots, f_K)$ producing predictions $(\hat{\mathbf{y}}_1, \ldots, \hat{\mathbf{y}}_K)$ on the input matrix $(\mathbf{X})$, a meta-learner (g) is trained to combine these outputs, as depicted below:

$$\hat{\mathbf{y}}_{\text{final}} = g(\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2, \ldots, \hat{\mathbf{y}}_K) \tag{43}$$

The meta-learner, therefore, learns the optimal method to combine the base model predictions to produce the final outcome.

Reference: Wolpert (1992).

0.6.5 Reliability Analysis and Evaluation

Domain shift assessment Domain shift refers to discrepancies between training and future test datasets, impacting model generalisation (Gama et al., 2014). Four primary types of domain shift are examined.

Data Drift. Data drift occurs when the distribution of the training dataset differs from future test samples (Gama et al., 2014). This is assessed using:

- **Kolmogorov-Smirnov (K-S) test**: Measures the largest difference between empirical cumulative distributions (Kolmogorov, 1933; Smirnov, 1948).
- **Drift Score Index (DSI)**: Quantifies divergence in statistical moments. While commonly used in industry for monitoring, a single foundational academic paper for DSI specifically is less common than for other statistical tests.
- **Wasserstein Distance (WD) test**: Computes the optimal transport distance between distributions, offering a more robust measure for comparing distributions, particularly for continuous data (Arjovsky et al., 2017).

$$D_{\text{KS}} = \sup_x |F_{\text{train}}(x) - F_{\text{test}}(x)| \tag{44}$$

Figure 22: Conceptual Diagram of Data Drift Assessment (Authors' conceptualisation)

Concept Drift. Concept drift occurs when the relationship between features and target variables shifts over time (Gama et al., 2014). It is evaluated using:

- **Kullback-Leibler (KL) Divergence**: Measures how one probability distribution differs from another, quantifying information loss when approximating one distribution with another (Kullback and Leibler, 1951).
- **Jensen-Shannon (JS) Divergence**: A smoothed, symmetric variation of KL divergence, often preferred for its boundedness and symmetry, making it a true metric (Lin, 1991).
- **Population Stability Index (PSI)**: Assesses stability in feature-target relationships, commonly used in credit risk and fraud detection to monitor changes in score distributions over time (Siddiqi, 2012).

$$D_{\text{KL}}(P|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)} \tag{45}$$

Figure 23: Conceptual Diagram of Concept Drift Assessment (Authors' conceptualisation)

Feature Drift. Feature drift measures the importance contained in features between training and test datasets. It is analyzed using:

- K-S test (referenced from Data Drift) (Kolmogorov, 1933).
- Population Stability Index (PSI) (Siddiqi, 2012).
- Chi-square test: Compares categorical feature distributions, assessing the independence between two categorical variables or the goodness-of-fit of a distribution (Pearson, 1900).

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \tag{46}$$

Figure 24: Conceptual Diagram of Feature Drift Assessment (Authors' conceptualisation)

Covariate Shift. Covariate shift evaluates the distributional shift in input variables between training and test sets (Shimodaira, 2000). Since standard formulas may not exist, a **pseudo-code** representation is provided.

Pseudo-code Presentation. Illustration of the Covariate Shift detection method.

```
FUNCTION CovariateShiftAssessment(X_train, X_test):

    INITIALIZE covariate shift detection metrics

    COMPUTE probability density estimates for each feature
    in X_train and X_test

    MEASURE statistical deviations between train and
    test distributions

    IDENTIFY significant shifts using pre-defined
    threshold parameters

    FLAG features with high divergence scores

    RETURN Covariate Shift Indicators
```

Figure 25: Covariate Shift Assessment Pseudo-code (Authors' conceptualisation)

Model Performance Evaluating model reliability requires assessing deterioration over time and biases affecting fairness. This section examines techniques for detecting these issues.

Model Deterioration. Performance degradation occurs when the accuracy of a trained model decreases over time due to evolving data distributions (Tsymbal, 2004). Two key techniques for deterioration assessment include:

- **Non-Parametric Bootstrap**: Estimates variation in model outputs over repeated sampling, providing confidence intervals for model stability (Efron, 1979).
- **Regression Analysis**: Models the trend of accuracy decay over time, quantifying systematic deterioration (Wooldridge, 2016).

Mathematical Definition. Given historical model performance (y t), deterioration can be approximated using:

$$y_t = \beta_0 + \beta_1 t + \epsilon_t \tag{47}$$

where:

- (0) is the initial performance baseline.
- (1) is the rate of deterioration over time.
- (t) represents residual errors.

Additionally, the Non-Parametric Bootstrap method estimates confidence intervals for model stability:

$$\hat{\theta}\text{bootstrap} = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}^{(b)} \tag{48}$$

where:

- (B) represents the number of bootstrap resampling iterations.
- ((b)) is the estimated parameter in the (b th) resample.

Pseudo-code Presentation. Illustration of the Bootstrap Resampling process.

```
FUNCTION BootstrapResampling(data, num_samples):

    INITIALIZE sample distribution
    FOR each bootstrap iteration b IN {1, ..., num_samples}:
        RESAMPLE data with replacement
        COMPUTE estimator value for resampled data
        STORE computed estimator in distribution

    RETURN bootstrap confidence intervals
```

Figure 26: Bootstrap Resampling Pseudo-code (Authors' conceptualisation)

Bias / Fairness Issues. Bias in model predictions can lead to unfair decision-making and disparate impacts across demographic groups (Barocas et al., 2017). Three statistical approaches are used:

- **Statistical Parity**: Ensures similar selection probabilities for different demographic groups (Dwork et al., 2012).
- **Equalized Odds**: Evaluates fairness across true positive and false positive rates between groups (Hardt et al., 2016).
- **Predictive Parity**: Ensures that predicted outcomes maintain similar probabilities across different groups (Chouldechova, 2017).

Mathematical Definition. Fairness techniques can be defined as:

$$\text{Statistical Parity} = P(\hat{Y} = 1|G = 0) \approx P(\hat{Y} = 1|G = 1) \tag{49}$$

$$\text{Equalized Odds}: \quad P(\hat{Y} = 1|Y = 1, G = 0) = P(\hat{Y} = 1|Y = 1, G = 1) \quad \text{and} \quad P(\hat{Y} = 1|Y = 0, G = 0) = P(\tag{50}$$

$$\text{Predictive Parity}: \quad P(Y = 1|\hat{Y} = 1, G = 0) = P(Y = 1|\hat{Y} = 1, G = 1) \tag{51}$$

where:

- (G) represents the protected demographic group.
- (Y-hat) is the predicted outcome.
- (Y) is the actual label.

Pseudo-code Presentation. Illustration of the Bias Fairness Evaluation process.

```
FUNCTION BiasFairnessEvaluation(y_true, y_pred, group):

    COMPUTE statistical parity: P(predicted = 1 | group 0)
    COMPARE with P(predicted = 1 | group 1)

    COMPUTE equalized odds: P(predicted = 1 | actual = 1, group 0)
    COMPARE with P(predicted = 1 | actual = 1, group 1)

    COMPUTE predictive parity: P(actual = 1 | predicted = 1, group 0)
    COMPARE with P(actual = 1 | predicted = 1, group 1)

    RETURN fairness evaluation scores
```

Figure 27: Bias Fairness Evaluation Pseudo-code (Authors' conceptualisation)

Model Explainability Model explainability ensures transparency in predictive models, allowing analysts to interpret decisions effectively. Three key techniques are examined: SHAP values, Local Interpretable Model-Agnostic Explanations (LIME), and Causal AI.

SHAP Values. SHapley Additive exPlanations (SHAP) leverage cooperative game theory to attribute feature importance in predictions (Lundberg and Lee, 2017). SHAP assigns a contribution score to each feature based on marginal impact.

Mathematical Definition. Given feature set (X), SHAP value (j) for feature (j) is computed as:

$$\phi_j = \sum_{S \subseteq X \setminus j} \frac{|S|!(|X| - |S| - 1)!}{|X|!} [f(S \cup j) - f(S)] \tag{52}$$

where:

- (f(S)) represents the model output for feature subset (S).
- (X) is the total number of features.

Figure 28: Conceptual Diagram of SHAP Value Calculation (Authors' conceptualisation)

LIME (Local Interpretable Model-Agnostic Explanations). LIME generates locally interpretable surrogate models for black-box predictions by perturbing input samples and analysing feature importance (Ribeiro et al., 2016).

Mathematical Definition. Given perturbed instance (x prime), LIME minimizes:

$$\mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{53}$$

where:

- (L) represents fidelity loss between original model (f) and surrogate (g).
- (x) assigns weight to perturbed samples.
- ((g)) penalizes complexity.

Figure 29: Conceptual Diagram of LIME Explanations (Authors' conceptualisation)

Causal AI. Causal AI enhances interpretability by uncovering causal dependencies within feature interactions (Pearl and Mackenzie, 2018). If mathematical formulations are unavailable, a **pseudo-code** representation is provided.

Pseudo-code Presentation. Illustration of the Causal AI explainability method.

```
FUNCTION CausalAIAnalysis(X, y):

    IDENTIFY dependencies among features using causal graphs

    COMPUTE direct and indirect causal effects

    ESTIMATE intervention impact on predictions

    VISUALIZE causal pathways for interpretability

    RETURN Feature Importance & Causal Insights
```

Figure 30: Causal AI Explainability Pseudo-code (Authors' conceptualisation)

Evaluation Metric Mismatch Traditional accuracy-based metrics may not fully reflect model performance. This section evaluates different metrics across dataset types.

Comparison of Evaluation Metrics. Evaluation methods include:

- Accuracy, Precision, Recall, F1-score, and Matthews Correlation (Tharwat, 2018).
- Confusion matrices for classification performance (Powers, 2011b).
- Statistical assessments such as ANOVA, MANOVA, and Chi-square tests (Tabachnick and Fidell, 2013; Pearson, 1900).

Mathematical Definition. Different evaluation metrics are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{54}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{55}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{56}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{57}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{58}$$

where:

- (TP) = True Positives
- (TN) = True Negatives
- (FP) = False Positives
- (FN) = False Negatives

Figure 31: Conceptual Diagram of a Confusion Matrix (Authors' conceptualisation)

Advanced Statistical Analysis. Beyond standard metrics, statistical tests are applied to analyze deeper model reliability:

- **ANOVA (Analysis of Variance)**: Evaluates differences in model performance across multiple experimental groups (Fisher, 1925).
- **MANOVA (Multivariate Analysis of Variance)**: Extends ANOVA by analysing multiple dependent variables simultaneously (Rencher, 2002).
- **Chi-square Test**: Assesses categorical variable dependencies and fairness in model predictions (Pearson, 1900).

Mathematical Definition. ANOVA and Chi-square tests are computed as:

$$F = \frac{\text{Between Group Variance}}{\text{Within Group Variance}} \tag{59}$$

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \tag{60}$$

where:

- (O i) = Observed frequency.
- (E i) = Expected frequency.

Pseudo-code Presentation. Illustration of evaluation metric selection for model performance.

```
FUNCTION ModelEvaluationMetrics(y_true, y_pred):

    COMPUTE accuracy, precision, recall, F1-score, MCC

    IF dataset is imbalanced:
        APPLY precision-recall analysis instead of accuracy

    IF categorical feature dependencies need validation:
        PERFORM Chi-square Test

    IF multiple features influence performance:
        APPLY ANOVA or MANOVA analysis

    RETURN Evaluation Summary
```

Figure 32: Evaluation Metric Analysis Pseudo-code (Authors' conceptualisation)

0.7 Ethical Considerations

This study prioritizes stringent ethical compliance throughout the acquisition, usage, and processing of structured, textual, and image-based datasets. The research design has been developed to adhere to the highest standards of data governance and responsible conduct.

0.7.1 Permission and Informed Consent

Formal permission will be proactively obtained from all relevant data custodians or media entities. Prior to any data engagement, comprehensive confidentiality agreements

and robust legal safeguards will be meticulously established to ensure data integrity and intellectual property protection.

0.7.2 Anonymity and Confidentiality

In strict adherence to the principles of data privacy and the Protection of Personal Information Act (POPIA), all customer or subject-related data, particularly from consumer datasets, will be thoroughly anonymized. Measures will be implemented to ensure that all data is securely stored and protected from any unauthorized access, use, or disclosure, thereby safeguarding the privacy of individuals and entities involved. As this research does not involve live human subjects, animals, or direct environmental intervention, the ethical risks associated with biological or ecological systems are not applicable; nevertheless, all data will be treated with utmost discretion.

0.7.3 Ethical Clearance and Compliance

Ethical clearance for this study will be formally requested from the North-West University (NWU) Research Ethics Committee. This process will be facilitated by the supervisory team through a comprehensive proposal defense and subsequent official approval by the NWU Scientific Committee. All subsequent analysis, documentation, and dissemination of research findings will strictly adhere to both institutional guidelines and national legislative frameworks for responsible research conduct and academic integrity.

0.8 Layout of the Study

This section outlines the structure of the research study, detailing the focus and key aspects of each chapter in the 5-chapter thesis format.

- **Chapter 1: Introduction**
  - Problem, aims, objectives, research questions, and scope.
- **Chapter 2: Literature Review**
  - Foundational theory, ML/meta-learning models, data partitioning, reliability analysis, and identified gaps.
- **Chapter 3: Methodology**
  - Detailed plan for datasets, preprocessing, models, partitioning strategies, and the overall reliability framework.
- **Chapter 4: Analysis and Results**

  - Exploratory Data Analysis (EDA), model training, performance benchmarking, and reliability checks.

- **Chapter 5: Conclusion & Recommendations**
  - Key findings, model recommendations, and future work.

0.9 Contribution of the Study

This study aims to make significant contributions to the field of statistical learning, classification methodology, and the advancement of reproducible and ethically aligned Machine Learning (ML) practices. These contributions include:

- The development of a novel Adaptive Train-Test Partitioning (ATTP) framework that integrates advanced statistical methods—specifically dimensionality reduction (LDA), unsupervised clustering (k-Means), and iterative optimisation—to produce statistically more representative and robust train-test splits for enhanced model stability and generalisability.
- Empirical Benchmarking Protocol: Empirical evidence derived from a rigorous, modality-aware benchmarking protocol that systematically compares the performance and statistical properties of Traditional ML Models (such as Logistic Regression, Random Forest, Naïve Bayes, SVM, CNN, and LSTM) against Meta-Learning Models (including Reptile, Stacking Meta-Learner, and Performance-Aware Meta-Selector) across diverse data modalities and under various data partitioning schemes.
- Comprehensive Reproducible Reliability Assessment Framework: The creation of a comprehensive and reproducible reliability assessment framework for classification models. This framework statistically quantifies critical aspects such as drift (employing metrics like the Kolmogorov-Smirnov Test and Population Stability Index), evaluates fairness deviations across sensitive subgroups (using indicators like Statistical Parity, Equalized Odds, and Predictive Parity), and enhances model explainability (through tools such as SHAP and Causality checks). This ensures predictive integrity and trustworthiness prior to model application.
- Statistical Advancement in Model Selection: A direct contribution to statistical practice through the explicit quantification of uncertainty and generalisation error during the model selection process. The study establishes a formal, statistically-grounded procedure where reliability metrics (drift, bias, fairness) are used as primary statistical criteria (beyond simple accuracy) to transition from a proposed framework to the final recommended model for each modality, thereby advancing the methodology for trustworthy AI deployment.
- The establishment of a robust statistical foundation for advancing interpretable, fair, and adaptable AI models within multimodal domains. This specifically caters

to applications where transparent and statistically sound classification outcomes are paramount, thereby extending the utility and trustworthiness of ML in scientific and and practical contexts.

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Appendix A

Appendix A - Public Use Compliance Records

Annexure A-1: Credit Approval Dataset

This record details the compliance information for the Credit Approval dataset, a widely-used resource for classification tasks.

- **Dataset Description and Characteristics:** This dataset concerns credit card applications, featuring a mix of categorical, integer, and real attributes, with some missing values. All attribute names and values are anonymized for confidentiality. It is a multivariate dataset with 690 instances and 15 features, including a categorical target variable 'A16' (representing '+' or '-'). It is primarily used for binary classification tasks.
- **Source Origin and Licensing Agreement:** The Credit Approval dataset is sourced from the UCI Machine Learning Repository, a prominent online archive for machine learning datasets.
- **Scope of Permissible Usage:** The dataset is licensed under a **Creative Commons Attribution 4.0 International (CC BY 4.0) license**. This license permits users to share (copy and redistribute the material in any medium or format) and adapt (remix, transform, and build upon the material for any purpose, even commercially), provided appropriate credit is given to the original creator.
- **Statement of De-identification or Absence of Sensitive Personal Data:** The dataset has undergone anonymization, where all attribute names and their corresponding values have been obscured to protect the privacy and confidentiality of individuals. As such, it does not contain any personally identifiable information (PII).
- **Licensing Link or DOI for Verification:**
  - **Dataset Link:** `https://archive.ics.uci.edu/dataset/27/credit+approval`
  - **Reference DOI:** `https://doi.org/10.24432/C5FS30`

Annexure A-2: Wine Quality - Red Dataset

This record outlines the compliance details for the Wine Quality - Red dataset, a valuable resource for studying the relationship between physicochemical properties and wine quality.

- **Dataset Description and Characteristics:** This dataset is a segment of the "Vinho Verde" wine collection. It contains physicochemical and sensory variables specifically for red wine. Key characteristics include 11 real-valued features (e.g., fixed acidity, volatile acidity, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol) with an ordered and often imbalanced output class for 'quality' (ranging from 3 to 8). It is suitable for both classification and regression tasks related to predicting wine quality.
- **Source Origin and Licensing Agreement:** The Wine Quality - Red dataset is sourced from the UCI Machine Learning Repository.
- **Scope of Permissible Usage:** The dataset is licensed under a **Creative Commons Attribution 4.0 International (CC BY 4.0) license**. This license allows for broad reuse, including sharing and adapting the material for any purpose (even commercial), with the stipulation that appropriate credit must be given to the original source.
- **Statement of De-identification or Absence of Sensitive Personal Data:** The dataset consists purely of scientific measurements related to wine composition. It does not contain any personally identifiable information (PII) or sensitive data from individuals.
- **Licensing Link or DOI for Verification:**
  - **Dataset Link:** `https://archive.ics.uci.edu/dataset/186/wine+quality`
  - **Direct Data Import Link:** `https://archive.ics.uci.edu/ml/machine-learning-database wine-quality/winequality-red.csv`
  - **Reference DOI:** `https://doi.org/10.24432/C56S3T`

Annexure A-3: Wine Quality - White Dataset

This record provides the compliance information for the Wine Quality - White dataset, used to analyze physicochemical properties and their impact on white wine quality.

- **Dataset Description and Characteristics:** As a complement to the red wine dataset, this collection includes physicochemical and sensory variables specifically for white wine. Similar to its red counterpart, it features 11 real-valued attributes

(e.g., fixed acidity, volatile acidity, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol) and an ordered, often imbalanced 'quality' output class (ranging from 3 to 9). It is widely used for classification or regression tasks aimed at predicting white wine quality.

- **Source Origin and Licensing Agreement:** The Wine Quality - White dataset is sourced from the UCI Machine Learning Repository.
- **Scope of Permissible Usage:** The dataset is licensed under a **Creative Commons Attribution 4.0 International (CC BY 4.0) license**. This license permits extensive reuse, including sharing and adapting the material for any purpose (even commercial), provided that appropriate attribution to the original source is given.
- **Statement of De-identification or Absence of Sensitive Personal Data:** The dataset exclusively contains scientific measurements pertaining to white wine composition. There is no inclusion of personally identifiable information (PII) or sensitive individual data.
- **Licensing Link or DOI for Verification:**
  - **Dataset Link:** `https://archive.ics.uci.edu/dataset/186/wine+quality`
  - **Direct Data Import Link:** `https://archive.ics.uci.edu/ml/machine-learning-database wine-quality/winequality-white.csv`
  - **Reference DOI:** `https://doi.org/10.24432/C56S3T`

Annexure A-4: Yahoo Finance (JSE) Dataset

This record details the compliance for the Yahoo Finance (JSE) dataset, used for time-series analysis and forecasting of South African stock market trends.

- **Dataset Description and Characteristics:** This dataset comprises historical stock closing prices for 12 South African listed companies, covering a significant period from December 1979 to September 2025. It is a multivariate time series dataset with approximately 3292 instances per stock, including original features (Date, Close, ticker) and enriched features related to historical economic shocks relevant to South Africa. It is typically used for predicting directional price changes over historical windows.
- **Source Origin and Licensing Agreement:** Historical stock data is accessed programmatically via the `yfinance` Python API, which directly interfaces with publicly available data from Yahoo Finance. The `yfinance` tool itself is distributed under the Apache Software License.
- **Scope of Permissible Usage:** While the `yfinance` tool permits research and educational use of its API, it is crucial to note that Yahoo!'s official terms of use generally

restrict the underlying data to **personal use only**. Any use beyond personal, non-commercial purposes requires careful review and compliance with Yahoo!'s specific terms of service.

- **Statement of De-identification or Absence of Sensitive Personal Data:** This dataset contains publicly traded financial market data (historical stock prices). It inherently does not contain any personally identifiable information (PII) of individuals.
- **Licensing Link or DOI for Verification:**
  - **Yahoo Finance Website:** `https://finance.yahoo.com` (for underlying data source)
  - `yfinance` **GitHub (API):** `https://github.com/ranaroussi/yfinance` (for the tool used to access data)

Annexure A-5: South African News Data

This record summarizes the compliance information for the South African News Data, a valuable resource for natural language processing, particularly in under-resourced African languages.

- **Dataset Description and Characteristics:** This dataset comprises news content extracted from South African sources, specifically public Facebook pages of SABC's Motsweding FM and Thobela FM. It is a text-based dataset primarily used for news classification tasks, providing insights into topic modeling and language understanding within a South African context.
- **Source Origin and Licensing Agreement:** The South African News Data is publicly available and distributed via Zenodo, a general-purpose open-access repository.
- **Scope of Permissible Usage:** The dataset is licensed under a **Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license**. This license permits users to share and adapt the material, provided appropriate credit is given and the use is strictly non-commercial. Commercial use requires separate permission from the copyright holder.
- **Statement of De-identification or Absence of Sensitive Personal Data:** The dataset consists of publicly posted news content from social media platforms. All content is sourced from public domains, and the dataset has been compiled without including any personally identifiable information (PII).
- **Licensing Link or DOI for Verification:**

  - **Public URL Link:** `https://zenodo.org/records/3668495`
  - **Reference DOI:** `https://doi.org/10.5281/zenodo.3668495`

Annexure A-6: Tunizi: Tunisian Arabizi Sentiment Analysis Dataset

This record provides compliance details for the Tunizi dataset, a crucial resource for sentiment analysis in the specific dialect of Tunisian Arabizi.

- **Dataset Description and Characteristics:** The Tunizi dataset is noted as the first 100% Tunisian Arabizi sentiment analysis dataset. It comprises 100,000 comments extracted from social media, which have been manually annotated with positive, negative, and neutral polarities. The data has been preprocessed to remove noise elements such as links, emojis, and punctuations. It is a text dataset with a balanced split for training, validation, and test sets (7:1:2 ratio), making it suitable for sentiment classification.
- **Source Origin and Licensing Agreement:** The Tunizi dataset is publicly available and distributed via Zenodo.
- **Scope of Permissible Usage:** The dataset is licensed under a **Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license**. This license allows for sharing and adaptation of the material for non-commercial purposes only, requiring appropriate attribution to the original creators. Commercial use is not permitted under this license.
- **Statement of De-identification or Absence of Sensitive Personal Data:** The dataset consists of anonymized social media comments. Prior preprocessing to remove links, emojis, and punctuations further contributes to the de-identification process, ensuring the absence of personally identifiable information (PII).
- **Licensing Link or DOI for Verification:**
  - **Public URL Link:** `https://zenodo.org/records/4275240`
  - **Reference DOI:** `https://doi.org/10.5281/zenodo.4275240`

Annexure A-7: Amazon Commerce Reviews Dataset

This record outlines the compliance information for the Amazon Commerce Reviews dataset, primarily used for authorship identification research.

- **Dataset Description and Characteristics:** This dataset is specifically designed for authorship identification through online writeprint analysis. It is derived from customer reviews posted on the Amazon Commerce Website and includes linguistic style features. It is characterized as multivariate, text-based, and domain-theory focused, with 1500 instances and 10000 features. The dataset targets identifying the 50 most active users, each contributing 30 reviews. It is suitable for clustering and classification tasks based on review content and associated metadata.
- **Source Origin and Licensing Agreement:** The Amazon Commerce Reviews dataset is sourced from the UCI Machine Learning Repository.
- **Scope of Permissible Usage:** The dataset is licensed under a **Creative Commons Attribution 4.0 International (CC BY 4.0) license**. This license permits extensive reuse, including sharing, adaptation, and commercial use, provided that appropriate credit is given to the original source.
- **Statement of De-identification or Absence of Sensitive Personal Data:** The dataset consists of publicly available customer reviews. While review content itself is present, the focus is on linguistic style features for authorship identification. The dataset does not contain personally identifiable information (PII) beyond the publicly posted review text.
- **Licensing Link or DOI for Verification:**
  - **Dataset Link:** `https://archive.ics.uci.edu/dataset/215/amazon+commerce+reviews+set`
  - **Reference DOI:** `https://doi.org/10.24432/C55C88`

Annexure A-8: MNIST (Image Classification) Dataset

This record provides compliance details for the MNIST dataset, a foundational benchmark in image classification and deep learning.

- **Dataset Description and Characteristics:** MNIST is a foundational dataset comprising 70,000 grayscale handwritten digits (0-9). Each image is size-normalized to 28x28 pixels. It is widely used for training and testing image classification models due to its simplicity, clear visual patterns, and balanced class distribution. The dataset is split into a 'train' set of 60,000 examples and a 'test' set of 10,000 examples, with each instance containing an image (28x28x1, uint8) and a corresponding label (int64, across 10 classes).
- **Source Origin and Licensing Agreement:** This dataset is commonly accessed via TensorFlow Datasets (TFDS), which provides a convenient interface to many public datasets. The original source is attributed to Yann LeCun, Corinna Cortes, and Christopher J.C. Burges.

- **Scope of Permissible Usage:** The dataset is licensed under a **Creative Commons Attribution 4.0 International (CC BY 4.0) license**. This license allows for broad use, including sharing and adaptation for any purpose (even commercial), provided that appropriate credit is given to the original creators.
- **Statement of De-identification or Absence of Sensitive Personal Data:** The dataset consists solely of images of handwritten digits. It contains no human subjects' data or any personally identifiable information (PII).
- **Licensing Link or DOI for Verification:**
  - **Dataset Link (TFDS):** `https://www.tensorflow.org/datasets/catalog/mnist`
  - **Reference (Original Source):** `http://yann.lecun.com/exdb/mnist`

Annexure A-9: Hands and Palm Images Dataset

This record outlines the compliance information for the Hands and Palm Images Dataset, a resource for biometric applications and image-based classification.

- **Dataset Description and Characteristics:** This dataset is a comprehensive collection of 11,076 hand images (1600 x 1200 pixels) from 190 subjects aged 18-75. It includes images of both right and left hands, from dorsal and palmar sides, captured against a white background. The dataset is accompanied by detailed metadata, making it suitable for image-based classification tasks such as gender and age detection, and other biometric applications.
- **Source Origin and Licensing Agreement:** This dataset is publicly available and primarily distributed via Kaggle.
- **Scope of Permissible Usage:** The dataset is licensed under the **Database Contents License (DbCL) v1.0**. This license grants a worldwide, royalty-free, non-exclusive, perpetual, irrevocable copyright license for the contents of the database, including permission for commercial use. However, users **must comply with the Open Database License (ODbL)** for database rights, which typically requires attribution, share-alike, and keeping the database open. The dataset is explicitly stated as "FREE for reasonable academic fair use."
- **Statement of De-identification or Absence of Sensitive Personal Data:** The dataset contains images of human hands and palms, which are biometric in nature. While the metadata includes age and gender, specific efforts for de-identification beyond the provided dataset structure are not explicitly detailed. Users are advised to adhere to ethical guidelines regarding biometric data, especially when used in academic contexts, and ensure compliance with relevant privacy regulations like

POPIA. For this study, the data is treated as de-identified given its public availability and intended academic use.

- **Licensing Link or DOI for Verification:**
  - **Dataset Link:** `https://www.kaggle.com/datasets/shyambhu/hands-and-palm-images-data resource=download`
  - **Reference DOI:** `https://doi.org/10.1007/s11042-019-7424-8`